IMPACT OF ENHANCED CONTEXT-AWARENESS FOR CONSTRUCTION FIELD CREWS ON TASK CYCLE TIME

A DISSERTATION
SUBMITTED TO THE DEPARTMENT OF
CIVIL AND ENVIRONMENTAL ENGINEERING
AND THE COMMITTEE ON GRADUATE STUDIES
OF STANFORD UNIVERSITY
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR
THE DEGREE OF
DOCTOR OF PHILOSOPHY

RUI LIU

DECEMBER, 2021

© 2021 by Rui Liu. All Rights Reserved.

Re-distributed by Stanford University under license with the author.

This work is licensed under a Creative Commons Attribution-Noncommercial 3.0 United States License.
http://creativecommons.org/licenses/by-nc/3.0/us/

This dissertation is online at: https://purl.stanford.edu/hk906xs6153

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

Martin Fischer, Primary Adviser

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

Oussama Khatib

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

Michael Lepech

I certify that I have read this dissertation and that, in my opinion, it is fully adequate in scope and quality as a dissertation for the degree of Doctor of Philosophy.

Ben Schwegler

Approved for the Stanford University Committee on Graduate Studies.

Stacey F. Bent, Vice Provost for Graduate Education

This signature page was generated electronically upon submission of this dissertation in electronic format. An original signed hard copy of the signature page is on file in University Archives.

Abstract

Lacking awareness of context in a timely manner has hampered field communication on construction jobsites. As a result, field crews are often not informed of the right information at the right time and the performance of field tasks may not be as good as necessary. This thesis provides a context taxonomy consisting of 17 context types and context-based work instructions that improve field communication by enhancing context-awareness. In addition, a case study shows that context-based work instructions improve field task performance by reducing the variability of cycle time.

Two research questions drove this research. The first question is (1) *How to define context and develop a context taxonomy for construction field crews so that they could be informed of context information and its changes?* A field study of 40 instances of field task problems on four construction jobsites showed that field crews were not sufficiently informed of context information and its changes. For example, a precast shear wall crew did not find out that the supporting brace was not of the right size until a shear wall was lifted to the installation location. Another example showed that a crew was notified to double the number of internet cables after the ceiling was installed. To address this question, the overall strategy was to clearly understand context for field crews and then I decided to explore context modeling methods to formally represent field context in a machine-interpretable way and support reasoning of context. First, therefore, I put forward an adjusted definition of field context based on a synthesis of previous efforts to define context. Second, a context taxonomy, consisting of 17 specific context types, was developed to provide a common vocabulary for describing field context. The 17 context types were validated through semi-structured interviews with 15 experienced industry professionals from sophisticated construction companies. Third, I built an ontology-based construction context information (CCI) model based on the validated context types. The CCI model defines relationships between these context types and their attributes. This research shows that, with the model, it is feasible to define reasoning rules to derive high-level context (e.g., a task's status) from low-level context (i.e., context that can be directly defined or sensed).

A case study on a precast parking structure project indicates that the cycle time of construction field tasks varies significantly due to crews' unawareness of context and its changes. I then wondered whether the cycle time of field tasks would be affected and, if so, how if field crews were provided with instructions incorporating field context information based on the CCI model. Hence, I asked the second research question: (2) *What is the impact of context-based work instructions on the cycle time of construction field tasks?* To answer this question, the overall strategy was to create context-based work instructions and design a controlled experiment to compare my proposed instructions with state-of-the-art instructions in terms of the impact on task cycle time. First, based on a review of features of high-quality work instructions, I decided to focus on two distinguishing features: informed of changes and just-in-time. By meeting these two distinguishing features, the context-based work instructions formalized in this research can help inform field crews of context information and its changes in a timely manner. Second, I created context-based work instructions, the implementation of which was designed to align with short cycle planning processes through a mobile-based prototype due to the need to communicate context information in a just-in-time way. To evaluate how context-based work instructions affect field tasks, I conducted a controlled experiment on an office renovation project. The controlled experiment was performed over 14 groups of field tasks involving 246 instances of field tasks. Each group was equally divided into a control sub-group and a treatment sub-group. For the control sub-group, field crews were provided with state-of-the-art work instructions, while for the treatment sub-group, context-based work instructions were implemented. For each instance of a field task, a percent deviation was calculated by dividing the delta between the actual and the planned cycle time by the planned cycle time. A small percent deviation indicates that the planned cycle time is close to the actual, suggesting that the plan reflected the actual situation. The calculations showed that, across the 14 groups of tasks, the average percent deviation of the treatment sub-groups was 35.7% less than that of the control sub-groups. Based on the calculations, I find context-based work instructions help reduce the variability of the cycle time of field tasks with respect to planned vs. actual cycle times. The case study results also show that the actual cycle time varies less from day to day for the treatment group vs. the control group. Together, these findings imply that context-based work instructions help avoid overruns of task cycle time.

Overall, this thesis contributes a context taxonomy consisting of 17 context types as a common vocabulary for describing field context and context-based work instructions that communicate context information in a just-in-time way and help inform field crews of changes in context. The case study showed that, with context-based work instructions, the performance of field tasks was enhanced in that the variability in the cycle time of field tasks was reduced. Future work should study the generality of the results by evaluating the formalized work instructions in other projects and field tasks.

Acknowledgement

The years that I have gone through to finish my PhD work were filled with excitement, hope, uncertainties, self-doubt, frustration and achievement. Throughout this challenging journey, I was lucky enough to receive a lot of help from the brilliant people, to name a few:

I would like to first thank my PhD advisor, Martin Fischer, for guiding me through the journey. Martin was like a beacon to me the time when I chose to become a researcher in the AEC industry. I really cherish the opportunity to conduct research under his supervision. He gave me insightful feedback when I had questions, and more importantly he always urged me to think deeply and thoroughly so that I could own my research. He also helped improve my skills in writing and presentation so that I could effectively communicate my research with others.

I really appreciate the help from my committee members. Thanks, Oussama Khatib, for the illuminating discussions that we had about my research. Thanks, Ben Schwegler, for always being available when I asked for help. I was lucky to do an internship under the supervision of Ben in Shanghai Disney Research Lab. Ben gave me enlightening advice on how to carry out my research. Thanks, Michael Lepech, for teaching the life cycle analysis course, which helped me develop a holistic view of analyzing research questions. Thanks, Rishee Jain, for his insightful perspectives which shed light on my work. I would like to thank, Yinyu Ye, for being my chair and an inspiration to me.

I also feel grateful for the support from Center for Infrastructure Facility Engineering (CIFE) for providing seed funding for my research. In addition, CIFE members, including Clark Pacific, Bouygues Construction, DPR, offered me access to construction jobsites. I had opportunities to perform semi-structured interviews with experienced industry professionals from these companies. Thanks to Ruldoph and Sletten (R&S), for letting me visit their sites. I also appreciate the chances to talk with professionals from R&S.

I sincerely thank my CIFE colleagues. Thanks to Forest, Tulika, Ashwin, Min, Yujin, Hesam, Bo, Amanda, Alissa, Yanping, Bochen, Junwen…for your conversations, insights and suggestions on my work. Specifically, Forest helped me a lot at the early stage of my research, by providing me with his insightful perspectives and connecting me to local labor unions. As my research group members, Tulika and Ashwin met me on a weekly basis and they provided a lot of feedback on my work.

Thanks to my friends for their company along this journey. When I needed relax and comfort, they were always willing to spend time with me. Thank you so much, Wentao, Cici, Yongqiang, Yingqi, Lin, Yang, Yimei, Miao, Qing, Mengyi, Lisa…for supporting me and keeping me posted of their life updates. I would like to give my special thanks to Daisy for her support and company. You were a treasure of my life.

Lastly, I thank my parents, Hong Liu and Surong Qiu, and other family members. They give me a lot of love that I could never return back.

Table of Contents

Chapter 1: Introduction *1*

1. Conceptual framework 1
2. Clarifications on scope of work 4
3. Contribution to knowledge 5
4. Structure of the dissertation 6

References 8

Chapter 2: Context Taxonomy for Construction Field Crews *9*

1. Introduction 9
2. Preliminary field study 11
3. Related work 12

3.1 Definition and features of context 13

3.2 Context-awareness on construction jobsites 18

4. Analysis of construction work and field context 19

4.1 Characteristics of construction field work 20

4.2 Features of field context 21

5. Context for construction field crews 22

5.1 Definition of the field context 22

5.2 Construction context taxonomy 23

6. Methodology for validating the context taxonomy 24

6.1 Interview design 24

6.2 Demographic information of interviewees 25

7. Interview results 26

7.1 Product context 26

7.2 Organization context 28

7.3 Process context 29

7.4 External factors context 31

7.5 Summary 34

8. Study of field problems using identified context types 34
9. Discussion 36
10. Conclusions 38

Acknowledgement 40

Appendix 41

References **43**

Chapter 3: Development of A Construction Context Information (CCI) Model for Construction Field Crews ***51***

1. Introduction **51**

2. Related work **54**

3. Context types for construction field crews **58**

3.1 Product context **58**

3.2 Organization context **58**

3.3 Process context **59**

3.4 External factors context **59**

4. The Construction Context Information (CCI) model **60**

4.1 A task scenario on construction jobsites **60**

4.2 The top-level context ontology structure **61**

4.3 Product ontology **63**

4.4 Organization ontology **64**

4.5 Process ontology **65**

4.6 External factors ontology **66**

4.7 Summary **68**

5. Context reasoning **68**

6. Conclusions **70**

Acknowledgement **71**

References **72**

Chapter 4: Context-based Work Instructions for Construction Field Tasks ***76***

1. Introduction **77**

2. Motivation **78**

3. Preliminary case study **79**

4. Related work **82**

4.1 Criteria for work instructions **82**

4.2 Change management for construction field work **85**

4.3 Just-in-time principles and Last Planner System **86**

4.4 Context-aware applications on construction jobsites **88**

4.5 Summary **91**

5. Research method **91**

5.1 Collection of the field context **91**

5.2 Context-based work instruction prototype (MVP) **92**

5.3 Conduct a field study **93**
6. Field study results **95**
7. Discussion **97**
8. Conclusions **99**
Acknowledgement **101**
Appendix **102**
References **103**
Chapter 5: Conclusion ***108***
1. Theoretical contributions **108**
2. Practical contributions **110**
3. Limitations **111**
4. Future work **112**

List of Figures

Figure 1. Key concepts and relations of this thesis 2
Figure 2. Key terms and relations in this thesis (updated) 4
Figure 3. A context taxonomy for construction field crews 24
Figure 4. Percentage of occurrence of each context type 36
Figure 5. Top-level context ontology structure 62
Figure 6. Relationships of context types 63
Figure 7. Product ontology 64
Figure 8. Organization ontology 65
Figure 9. Process ontology 66
Figure 10. External factors ontology 67
Figure 11. An example of existing advanced work instruction (Sourced from Clark Pacific) 79
Figure 12. Cycle times of observed shear wall installations on a precast project 80
Figure 13. Production cycle time of shear walls on the ground level 81
Figure 14. Site photo: installation of precast shear wall 81
Figure 15. Revised feedback processes caused by changes (Motawa et al., 2007) 86
Figure 16. Preparation and update of context-based work instruction in line with the Last Planner System processes 88
Figure 17. Sources for the 17 context types 92
Figure 18. Percent deviation of the control and treatment sub-groups 96

List of Tables

Table 1. Planned paper submission 7
Table 2. List of projects and instances of field task problems 12
Table 3. Demographic information of participants 26
Table 4. Mapping of field problems to context types 36
Table 5. Results of coding for the interview responses 41
Table 6. Examples of reasoning rules 70
Table 7. Variance of the actual cycle time per quantity in the control and treatment sub-groups 97
Table 8. Calculations of percent deviation and its difference for the control and treatment sub-groups 102

Chapter 1: Introduction

1. Conceptual framework

A study on the productivity of construction jobsites found that inadequate communication and unavailability of required information were two main reasons accounting for extended completion times (Hewage and Rawanpura, 2009). It implies that effective field communication is crucial to ensure that field crews avoid schedule overruns. I was wondering how to improve field communication so that field tasks could be successfully completed on schedule. To further understand the communication problem and evaluate its impact on project schedules, I conducted a case study on a precast parking structure project, during which I used the cycle times of field tasks to measure schedule performance. The case study showed that the cycle time of installing the same precast component (e.g., shear wall) varied significantly from shear wall to shear wall. According to my records of communication problems faced by field crews, lack of awareness of context information and its changes on the jobsite was a significant reason for this variation. For example, the installation of a precast shear wall was delayed because the field crew spent extra time on preparing more grouting material than expected. It struck to me that field crews might not always have access to context information and its changes. My intuition was supported by evidence that there is no context taxonomy for construction field crews in prior studies. Therefore, to inform field crews of context information and its changes, the first step in this research had to be to develop a clear understanding of context for construction field crews. Abowd et al. (1999) defined context as any information that can be used to characterize the situation of an entity. One take-away from this definition is that recognizing the entity should be the first step to identify context information. In this thesis, the entity refers to field crews. The reasons are as follows. Value stream theory implies that construction field crew members are the ones who directly add value onto the final building product. Thus, how they perform field tasks and what they deliver determines the project performance. In addition, the above-mentioned study by Hewage and Rawanpura (2009) showed that poor communication with field crews causes schedule overruns. With the introduction of context, I was curious about how being informed of context would

impact the task cycle time for field crews. Figure 1 shows the key concepts and their relationships so far. Still, there is a question about what context information is transmitted to field crews and how. Moreover, the impact on task cycle time of providing field crews with timely informing of context information and its changes is unknown.

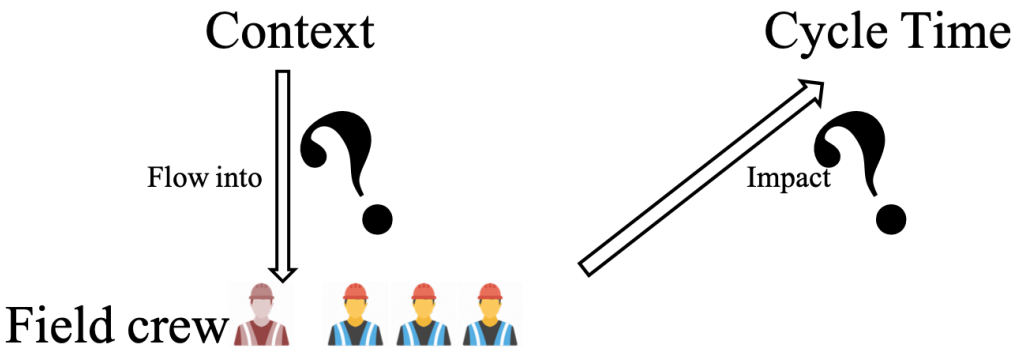

Figure 1. Key concepts and relations of this thesis

Previous studies show that although there have been attempts to increase awareness of context on construction jobsites (Singhvi and Terk, 2011; Lee and Mohsen, 2008), there is no common definition nor taxonomy for describing the context for field crews yet. As pointed out by Perera et al. (2013), a common definition of context is almost impossible since no definition applies to all domains. In other words, context is a domain-specific concept. Hence, adjustments to previous definitions (including Abowd's) should be made to put forward a definition applicable to construction field crews. it would be helpful for practical implementation to develop a context taxonomy so that field crews could have a common vocabulary for communicating context information. My analysis of 40 instances of field task problems on four jobsites showed that these problems are associated with repeated types of context (e.g., material, design). Thereafter, I was wondering whether the context taxonomy consists of a limited number of context types; if so, field crews may find it easy to recognize the context information they require.

With an adjusted definition of context for field crews, I referred to existing construction information models, mainly including the POP (Production, Organization and Process)

model, the AROW (Activity, Resource, Object, and Work Area) model and the AFM (Activity Flow Model), to come up with specific context types (Kunz and Fischer, 2012; Mourgues and Fischer, 2008; Garcia-Lopez and Fischer, 2016). After an initial list of context types was obtained, I validated whether these context types are of interest to field crews. Semi-structured interviews with experienced industry professionals from sophisticated construction companies were conducted for the validation. Since the interviewees were very experienced in working with field crews, they could provide objective opinions on the importance of each context type and whether a context type was missing in the initial set of context types.

Moreover, different context types may affect a crew simultaneously and therefore it is worthwhile to integrate the context information on construction jobsites. Similar to a language needing grammar to connect individual words to generate stronger expressiveness, it needs to be possible to appropriately relate the context types to each other to describe the situation faced by field crews when performing tasks on jobsites. Hence, I compared different context modeling methods. The ontology-based context modeling method has been shown to have advantages in expressiveness and inference capability over other modeling methods. Therefore, it was selected in this thesis to model context for field crews. The model was expected to describe how a relationship between two context types and attributes associated with each context type could be defined, but it was beyond the scope of this research to completely list the relationships and attributes.

Work instructions are the media for conveying information to field crews. Mourgues and Fischer (2008) developed a field work instruction template, whereas their instructions cannot respond to changes and provide the information in a just-in-time way. Hence, I decided to focus on two distinguishing features: informed of changes and just-in-time. Since context is dynamic and characterized by timeliness, work instructions that are developed based on context must address these two features. Therefore, I was curious about whether context-based work instructions could improve project schedule performance by reducing variations in the cycle time of field tasks. To answer this question, a controlled experiment was designed in which the same type of crew was provided state-of-the-art

work instructions in the control group while the crew in the treatment crew received context-based work instructions to perform the same type of task on the same project. If the task cycle time varied less in the treatment group than in the control group, it would show that context-based work instructions help reduce variations in task cycle time. Figure 2 updates Figure 1 by adding work instructions into the roadmap. Context information needs to be collected and used to developed work instructions, which will be further used to inform field crews of context information and its changes.

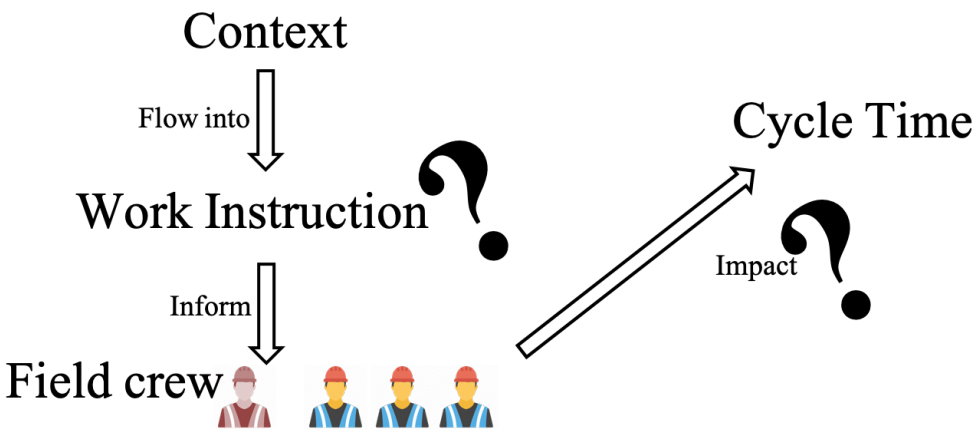

Figure 2. Key terms and relations in this thesis (updated)

2. Clarifications on scope of work

While field communication involves different participating actors, this thesis mainly focuses on the communication with construction field crews. Construction field crews are often not informed of context information and its changes in a timely manner, which could lead to dramatic variation of the cycle time of field tasks. To approach this communication problem, this research makes the following assumptions:

- There are two categorization schemes: operational categorization and conceptual categorization. Operational categorization considers how context is obtained and as such the context is categorized into sensed, static, profiled, and derived context. Conceptual categorization considers the meaning and conceptual relationships between context and exemplified context types. This thesis applies the conceptual

categorization schema and assumes that there are common context types. If there was an exhaustive list of common context types, field crews should be informed of these context types.

- Work instructions are the media for transmitting information with construction field crews. However, existing work instructions do not include context information and its changes in a just-in-time way. Hence, this thesis focuses on two distinguishing features of work instructions: informed of changes and just-in-time. If work instructions were developed by incorporating context information, it might help resolve this communication problem. Therefore, this research attempted to figure out whether context-based work instructions could help reduce the variability of the cycle time of field tasks.
- Project planning consists of long-term planning (e.g., master planning) and short-term planning (e.g., weekly planning). This thesis considers that the implementation of context-based work instructions should align with short-term planning. The reason is that context information should be specific enough to guide field crews to perform tasks.

3. Contribution to knowledge

This thesis makes the following contributions to knowledge:

- Based on a synthesis of previous efforts to define context, I put forward an adjusted definition of context specific to construction field work and developed a context taxonomy consisting of 17 field context types that provide a common language to describe the context information required by construction field crews when performing field tasks. Through semi-structured interviews with experienced industry professionals, this thesis validates these 17 context types and shows that these interviewees could not think of another context type to be included in the taxonomy. Hence, for researchers who are interested in the implementation of context in the construction industry, this thesis lays the foundation by providing a validated context taxonomy. In addition, this thesis shows that a construction context information (CCI) model was built upon the context taxonomy by defining

relationships between these context types and their attributes. The CCI model enhances the context taxonomy by improving the ability to interpret, reason about and communicate field context information.

- State-of-the-art work instructions do not include context information and its changes in a timely manner. This thesis suggests context-based work instructions that meet two distinguishing criteria: informed of changes and communicated in a just-in-time manner.
- Quantitative assessment of the impact of context-based work instructions was performed in a case study. The instructions were shown to help reduce the variability of the cycle time of field tasks. As implied by the case study result, context-based work instructions make the actual cycle time of field tasks more consistent with the planned cycle time and the actual cycle times vary less from day to day.

4. Structure of the dissertation

The dissertation applies the three-paper structure. Chapter 2, 3, and 4 are stand-alone papers that are ready for publication in academic journals. Table 1 lists the proposed paper titles, authors, and journals.

Table 1. Planned paper submission

Chapter	Proposed paper titles, authors and journals
Chapter 2	Rui Liu, Martin Fischer. "Context Taxonomy for Construction Field Crews". Manuscript planned for submission to Journal of Construction Engineering and Management.
Chapter 3	Rui Liu, Martin Fischer. "Development of A Construction Context Information (CCI) Model for Construction Field Crews". Manuscript planned for submission to Automation in Construction.
Chapter 4	Rui Liu, Martin Fischer. "Context-based Work Instructions for Construction Field Tasks". Manuscript planned for submission to Journal of Construction Engineering and Management.

References

Abowd, G.D., Dey, A.K., Brown, P.J., Davies, N., Smith, M. and Steggles, P., 1999, September. Towards a better understanding of context and context-awareness. In *International symposium on handheld and ubiquitous computing* (pp. 304-307). Springer, Berlin, Heidelberg.

Garcia-Lopez, N.P. and Fischer, M., 2016. A construction workflow model for analyzing the impact of in-project variability. In *Construction Research Congress 2016* (pp. 1998-2007).

Kunz, J. and Fischer, M., 2012. Virtual design and construction: themes, case studies and implementation suggestions. *Center for Integrated Facility Engineering, Stanford University*.

Lee, S. and Mohsen, O. (2008). "Application of context-aware computing to construction field work." *Proceedings of the CSCE 2008 Annual Conference*, Vol. 1, 483–490.

Mourgues, C. and Fischer, M., 2008. A work instruction template for cast-in-place concrete construction laborers. *Center for Integrated Facility Engineering. Stanford University. Working Paper*, *109*.

Hewage, K.N. and Ruwanpura, J.Y., 2009. A novel solution for construction on-site communication–the information booth. *Canadian Journal of Civil Engineering*, *36*(4), pp.659-671.

Perera, C., Zaslavsky, A., Christen, P. and Georgakopoulos, D., 2013. Context aware computing for the internet of things: A survey. *IEEE communications surveys & tutorials*, *16*(1), pp.414-454.

Singhvi, V. and Terk, M. (2011). "Prophet: a contextual information system framework." *CIB REPORT*, Vol. 284, 318.

Chapter 2: Context Taxonomy for Construction Field Crews

Abstract: Construction field crews can benefit from being informed of context, which refers to the relevant information for them to perform their tasks and changes of context in a timely manner. A study of 40 unique instances of field task problems on four construction jobsites showed that the inability to do so could cause an increase in task cycle time and sometimes safety incidents. However, there is currently no shared understanding of context for construction field crews, which makes the communication of context information and its changes difficult. In this study, a definition of context for construction field crews and a context taxonomy consisting of 17 different context types were developed. Semi-structured interviews with industry professionals were conducted to validate the context types. The context taxonomy provides a formal representation of context for construction field crews, and, thus, can provide a shared understanding of context and facilitate relevant research in the construction industry. For practical implementation, the context taxonomy can be used to identify context and its changes for construction field crews.

Key words: Context; Construction; Field Work; Crew Members; Taxonomy; Validation

1. Introduction

According to value stream mapping (VSM) theory, construction field crews are the ones who directly add value onto the final building product (Arbulu et al. 2003). Hence, project performance is largely dependent on how field crews perform their tasks and what they deliver. Effective communication is fundamental, but a study of 40 non-repetitive instances of field task problems on four different projects indicates that poor communication with field crews is still a common problem. An awareness of context is critical for good communication (Schilit et al., 2002). Context, until this point, can be simply understood as relevant information to construction field crews in this paper. Timely communication of context information and its changes could help field crews get the information needed for performing field tasks and deliver the building product that satisfies the client's requirements.

Since the construction environment is complex and dynamic, it is challenging to inform field crews of context information and its changes in a timely manner. Previous research has explored enabling technologies to increase context-awareness on construction jobsites. Aziz et al. (2005) reviewed enabling technologies for intelligent wireless web services to help field crews access context-relevant and personalized information. Rebolj et al. (2008) argue that reliable context information is critical for successful completion of tasks, and they propose a method consisting of three components: an automated activity tracking subsystem based on image recognition, an automated material tracking subsystem, and a mobile communication environment. Chen and Kamara (2011) introduce a framework consisting of an application model and a technical model. The application model identifies the features of mobile computing, construction personnel, construction information, and construction sites and explores the interactions affecting the implementation of mobile computing. The technical model presents a clear structure for designing mobile computing systems from a technical perspective. These research efforts recognized the importance of context information and were devoted to exploring technologies that could be leveraged to help field crews. However, we found that a clear understanding of context for field crews, which should be the foundation for utilizing context information, is missing.

A taxonomy could provide a common understanding of field context. Without being informed of context information and its changes, construction field crews could be confronted with various problems (e.g., delay, rework, finished work with a quality lower than expected). Consider a simple scenario when a field crew is assigned to install a ceiling while the members are not aware of a change to the number of internet cables above before starting the task. The ceiling has to be taken down afterwards, and the ceiling crew will have to perform rework. Such a scenario is common on construction jobsites. However, if the ceiling crew had been informed about the change in cables, its schedule could have been adjusted accordingly before they started.

This paper contributes an empirically validated taxonomy of context for construction field crews. The organization of this paper is as follows. Firstly, this study explored the meaning of context for construction field crews through literature review. From this review, an

adjusted definition of context for field crews and a context taxonomy consisting of 17 context types was developed. Secondly, the context taxonomy was validated through semi-structured interviews with 15 industry professionals. Thereafter, this study applied the taxonomy to a case project to evaluate whether it is able to describe real scenarios. The paper concludes by outlining how the context taxonomy could facilitate context-related research in the construction industry and, in practice, help inform construction field crews of context information and its changes in a timely manner.

2. Preliminary field study

To gain an initial understanding of the importance of context for construction field work, field studies on four different types of building projects were conducted and 40 unique instances of field task problems were observed. Each site visit lasted for 4 weeks during the construction stage. These four projects include a precast parking structure, a mixed-use commercial and residential building, an office building, and a medical facility.

Table 2 lists the four projects and the 40 instances of field task problems. For example, for the installation of a precast wall in project 1, field crews were confronted with the problem that a wrong wall was sent to the site, and crew members were not aware of this ahead of time. As a result, the crew needed to postpone the task even though they had already started mobilization. We made a first attempt at categorizing the problems to see whether there are common context types across different projects. Our categorization shows that the 40 instances of problems fall under 11 types including design (8), weather (7), material (5), work zone (5), equipment and tools (5), component (4), crew member (2), crew composition (1), public relation (1), technical and safety specification (1), and schedule (1). Common types of context (e.g., design, weather) that affected the execution of construction field tasks were found. Therefore, the analysis of these 40 instances of field task problems led us to a preliminary list of 11 context types. Moving forward, we wondered whether it is possible to define a set of context types that forms a vocabulary for describing context for construction field tasks.

Table 2. List of projects and instances of field task problems

#	Project	Problem Description	Field Task	#	Project	Problem Description	Field Task
1	**P1**	Wrong wall sent to site	Install precast wall	21		Unexpected share of crane	Rebar mobilization
2		No space for supporting brace	Install precast wall	22		Rain	Place concrete column
3		Precast wall needs adjustment of rebar angle	Install precast wall	23		Wrong rebar size	Install concrete column rebar
4		Precast wall needs cut of rebar	Install precast wall	24		Extract rainwater	Pour concrete column
5		Lack of cups	Install precast wall	25		first piece of arm is delayed	The third crane installation
6		No working space	Install precast wall	26		Concrete wall height and rebar frame height does not align	Place concrete column
7		Rainwater in cups	Install precast wall	27	**P3**	No space due to existing equipment	Install rotating door
8		Wrong supporting bracing	Install precast wall	28		Change location	Place concrete footing
9		Lack grouting material	Install precast wall	29		Concrete floor holes are closed	Place duct
10		Heavy rain	Install precast wall	30		Safety hazard	Mobilizing over opening area
11		Nearby column taking space	Install precast column	31		Unexpected change of design	Place sliding door
12		Equipment out of power	Install precast column	32	**P4**	Unintended gap between bottom of truss and window	Install truss over window
13		field crews idling	Install precast column	33		Double the amount of wire after ceiling is placed	Place wire over ceiling
14		Truck delivery delayed	Install precast column	34		Misalignment	Place drywall alongside window frame
15		Heavy rain	Install precast column	35		Height is not enough for installation	Install drywall in staircase area
16	**P2**	Rebar height does not match wall height	Cast-in-place concrete column	36		Cold weather	Place concrete
17		Supply is late	Install precast wall	37		Short of plumbers	Level 6 framing
18		Water extraction due to rain	Earthwork	38		Missing holes	Install fabricated steel floor beams
19		Muddy ground reduces speed	Moving materials	39		Dimensions are incorrect	Install elevator doors
20		Extract water into public pipes	Rainwater extraction	40		Need qualified plumbers	5th floor Plumbing issues with plumbers not tested

3. Related work

As argued by Perera et al. (2013), a common definition of context is almost impossible since no definition would be applicable to all domains. In other words, the meaning of context is domain-specific. For construction field crews, therefore, context should be

defined based on careful consideration of the features of construction field work. In order to build a shared understanding of context for construction field crews, we reviewed related research regarding (1) the definition of context in general and (2) context-awareness in the construction industry.

3.1 Definition and features of context

Context has been defined in four ways: 1) by enumerating examples; 2) by providing synonyms; 3) by categorizing information of an entity; or 4) by analyzing the constraints for an entity. In addition, we reviewed features of context for a deeper understanding of context.

3.1.1 Definition by Examples

According to Schilit et al. (1994), who first introduced the concept of "context-aware", context provides information about the located objects, people, and services in the environment and the changes to them. Some researchers enumerate examples such as location, identity, time, temperature, noise, as well as the beliefs, desires, commitments, and intentions of human beings (Brown et al., 1997; Gross and Specht, 2001; Ryan et al., 1998). These definitions are straightforward but may omit information that is of interest; for example, in this study, equipment or tools used by field crews are important but not included in the enumeration. From the implementation perspective, it is difficult to discern whether a certain type of information should be considered as context when the information cannot be categorized to a listed type.

3.1.2 Definition by Synonyms

Context is also defined using synonyms. For example, context is considered to have similar meaning to environment, situation, or circumstance (Brown, 1995; Ward et al., 1997). Unlike enumeration of examples, these definitions are too general to be used to determine what information should be considered as context. Since different people may have different understanding of the environment, situation, or circumstances, these definitions

define a term using another ambiguous term. In addition, given these terms with similar meaning, any definitions that include one of these terms will be self-referencing in loops (Zimmermann et al. 2007).

3.1.3 Definition by Categorization of Context Types

Another way to define context is by categorization. A categorization can help identify and structure context information, and as such can avoid being incomplete or too general. Van Bunningen et al. (2005) classify context categorization schemes into two types: operational categorization and conceptual categorization. The operational categorization scheme is based on how context is acquired, modelled, and treated; for example, Henricksen (2003) categorized context into sensed, static, profiled, and derived context. Conceptual categorization categorizes context based on the meaning and conceptual relationships between context. Based on the preliminary insights from the four case studies, we found it more urgent to understand context for field crews conceptually other than how to acquire or treat context. Hence, this study paid particular attention to research on conceptual categorizations of context.

Schilit et al. (1994) list the important aspects of context as where you are, who you are with, and what resources are nearby. Pascoe (1998) describe context as “the subset of physical and conceptual states of interest to a particular entity". Although he did not suggest specific context types, his definition indicates that context should be associated with an entity. For different entities, the meaning of context can be different since the information of interest could change. Ryan et al. (1998) argue that context should include both the physical and logical attributes of the environment. Abowd et al. (1999) define context as “any information that can be used to characterize the situation of an entity. An entity is a person, place, or object that is considered relevant to the interaction between a user and an application, including the user and application themselves”. However, this definition is still generic because it loosely limits the information that should be considered by using the term “any information”. To make the definition of context more explicit, they also suggest primary context types as “location, identity, activity, and time” to answer the questions of where, who, what, and when, and “use this information to determine why the situation is

occurring". Similarly, Abowd and Mynatt (2000a) identify the "five W's" (i.e., Who, What, Where, When, Why) to be the minimum set for context. Zimmermann et al. (2007) divide an entity's context into five categories: individuality, activity, location, time, and relations. According to their categorization, "the individuality category contains properties and attributes describing the entity itself. The category activity covers all tasks this entity may be involved in. The context categories location and time provide the spatio-temporal coordinates of the respective entity. Finally, the relations category represents information about any possible relation the entity may establish with another entity". Rosenberger et al. (2018) realized that a proper discussion of context in the industrial domain was lacking. Therefore, they modified Abowd's definition and added two distinctions: one is that context should consider the entity itself, and the other is that context should also contain immaterial information other than material objects. This study synthesizes these previous conceptual categorization schemes to develop context types (e.g., work zone) for construction field crews.

Moreover, Winograd (2000) built upon Abowd's definition by arguing that whether a certain type of information should be considered as context is based on assessing how it is used and its inherent properties. Truong et al. (2005) revised Abowd's definition by placing more emphasis on the role of context in interaction. Their definition considers context as a set of directions and information that define the interaction between the user and the application. More recently, Snidaro et al. (2015) considered context as the information that "surrounds" a situation of interest in the world. It is information that aids in understanding the situation and aids in reacting to the situation if a reaction is required. Based on the definitions by Truong et al. (2005) and Snidaro et al. (2015), context should consist of information that can help users to react to a certain situation. These studies indicate that field context should be directive and should help field crews react to the construction environment.

Overall, the definitions discussed above are attempts to categorize context into different types that are associated with an entity. Categorization of context types could serve as a taxonomy, and previous conceptual categorization schemes shed light on how context for

construction field crews could be categorized. A categorization of field context could aid field crews in accessing context information and its changes. However, previous categorization schemes might not remain applicable to construction field crews without adjustment because these schemes were not developed considering the construction environment.

3.1.4 Definition by Analysis of Constraint Relationship

Lastly, context has been defined from the perspective of analyzing constraint relationships. Rosenberger et al. (2005) describe context as "a feature F is contextual for an action A if F constrains A and may affect the outcome of A, but is not a constituent of A. Particularly, a situation, environment, or set of circumstances C is said to be a context for an action A if C constrains A or affects the outcome of A but is not a constituent of A". Their definition of context is applied to an action rather than an entity, whereas it emphasizes that context should be a set of constraints or have an impact on the outcome of the action. In fact, since an action should be executed by an entity (e.g., a person or an object), this definition becomes applicable to an entity when it is associated with an action. Compared to the definitions by categorization of context types, this definition can help to identify information that is of interest to an entity.

3.1.5 Features of Context

The definitions of context discussed above mostly are made from a static perspective, which means that these definitions barely consider changes of context; however, context is inevitably subject to uncertain changes (Henricksen and Indulska, 2004). Adomavicius and Tuzilin (2011) differentiate context into two dimensions: contextual variables and changing contextual variables. This differentiation indicates that recognizing uncertain changes is as important as identifying contextual variables themselves. Henricksen et al. (2002) identify imperfection as a major characteristic of context to indicate that context information can be incorrect or inconsistent.

Hence, to appropriately utilize context, it is necessary to account for uncertain changes and imperfections of context. Some researchers investigated quality metrics associated with context. For example, Lei et al. (2002) used freshness and confidence to measure how certain context is. Gray and Salber (2001) applied six quality metrics which are coverage, resolution, accuracy, repeatability, frequency, and timeliness. Gray and Salber (2004) propose three quality metrics describing the uncertainty of context: resolution, confidence, and freshness. According to these definitions of quality metrics, we found that different terms are used to measure some common dimension of the uncertainty of context; for example, freshness and timeliness both measure whether context information is up-to-date. To handle the uncertainty of context, Dey (2001) suggests a practical mechanism whereby users of context information are involved in a mediation process to validate the context. This solution is especially suitable for processing a small amount of context information when change is not rapid, so that the user is not unreasonably burdened (Bettini et al. 2010). These metrics along with Dey's mechanism inspired us to pay attention to changes of context for construction field crews. To satisfy the quality metrics listed above, field context should be dynamically checked and communicated to crews in a timely manner. At the start of this research, it was unknown how to do so, though.

3.1.6 Summary

To summarize, context should be closely associated with an entity. An entity can be a person, a place, or an object. The entity in this research is selected to be construction field crews. Furthermore, context should be information that is of interest to the entity. When there is some action taken by this entity, context should have impact on the process and the outcome of this action. To make the definition more specific and applicable, it is helpful to categorize context into different types and form a taxonomy. Lastly, it should be noted that context is subject to change over time and could be imperfect. Hence, any identified context information should be checked whether it is correct and should be communicated to field crews in a timely manner.

3.2 Context-awareness on construction jobsites

In the construction field, many researchers have attempted to increase context awareness. Singhvi and Terk (2011) developed a context information system framework named Prophet to proactively identify and retrieve information for workers on construction jobsites. They defined context as "any information that helps in the dynamic characterization of the construction environment. The basic elements of context are identity of the user, temporal information, location and community". However, their definition of context is not specific enough to guide field crews to perform tasks; for instance, field crews would not know what material and equipment to use. Lee and Mohsen (2008) developed a construction context classification and a Context Aware Computing (CAC) system to deliver appropriate information that can help workers deal with the dynamics of construction. In their research, the construction context classification includes work item context, user context, and environment context. However, the classification is not sufficient; for example, it does not consider how the work is performed. Wang et al. (2011) define context as "situations in which a construction entity acts or exists; whereby an entity can be a person involved in the construction, an action taken to complete a job, a component built in a project, or a resource utilized to perform the action or build the component". Further, they present a framework based on the notion of context descriptors. The framework is made up of two components: a representation of context descriptors using OWL-based ontologies and a reasoning mechanism to determine new descriptors' applicability when the number of context descriptors increases. Lee et al. (2013) developed a framework for Information Lifecycle Management (ILM) using RFID to help material control. They consider context as "on-going variables resulting from a given environment as well as a variety of participants, equipment, and scheduled works". Context parameters are identified as RFID checkpoint, user, time, location, and lifecycle stage. Despite these efforts to make use of context in the construction field, a consensus on the definition of context for construction field crews has not yet been reached.

Since context is a domain-specific concept, this study further reviewed research on existing construction knowledge models which could help identify the information of interest for

field crews. More specifically, with the need to conceptually categorize context for field crews, we referred to existing categorization schemes in the construction domain. Kunz and Fischer (2012) discuss the use of POP (Product, Organization, and Process) models to represent and manage information of a construction project. POP models complement 3D product models with process and organization models to become a more comprehensive method for Virtual Design and Construction (VDC). POP models are inter-connected and changes in one model often lead to changes in the other two. Moreover, Mourgues and Fischer (2008) developed the AROW (Activity, Resource, Object, and Work area) model to represent the information needed by field crews. Based on the AROW model, Mourgues and Fischer further developed a field instruction template to deliver task-specific information to field crews. The field instruction template contains different information sections such as work area, drawing, task procedures, bill of materials (BOM), and equipment and tools. Nevertheless, the instruction template lacks information (e.g., safety, surrounding environment conditions) and does not consider changes of the information. Garcia-Lopez and Fischer (2016) developed the AFM (Activity Flow Model) to represent flows between construction activities and account for the mechanisms causing workflow variability. The AFM model includes the following flows: labor, equipment, workspace, information and design, components, external conditions, and prerequisite work. The AFM is more detailed compared to POP and AROW, but it still neglects information such as safety.

The POP, AROW, and AFM models are referred to in this paper as existing categorization schemes of the information of interest for construction field crews at different levels of detail. Therefore, this study builds upon these existing construction knowledge models to identify an initial list of context types for field crews.

4. Analysis of construction work and field context

The review of previous studies on definitions and features of context has shown that context is a domain-specific concept. Therefore, a definition and a taxonomy of field context should accommodate the construction field environment. In this section,

characteristics of construction field work are discussed. Following is an analysis of the features of field context.

4.1 Characteristics of construction field work

To understand context in the setting of the construction field environment, it is necessary to consider the characteristics of construction field work. The construction environment is complex and dynamic (Park and Cho, 2017). The complexity can be attributed to factors including the environment, the use of a variety of resources, the interactions between different crews, and the level of scientific knowledge required (Dubois and Gadde 2002; Gidado, 1996). In addition, throughout the whole construction period of a project, various changes may arise and could affect the project schedule and costs (Thomas and Napolitan, 1995). Therefore, construction field work, in general, has the following characteristics.

- **A high volume of information:** Construction field crews receive large amounts of information on construction jobsites (Mak, 2001). They do not only need to appropriately utilize the information (e.g., schedules, drawings, safety requirements) to complete their tasks, they also need to be cautious of any changes of the given information, for example, changes of schedule.
- **Unstructured environment:** A construction job site is a temporary workspace. The environment is usually unstructured. There are, for example, stacks of materials and tools, presence of heavy equipment, possible existence of uncovered openings and hazardous materials (Guo, 2002).
- **High volatility:** Project plans and schedules are prone to changes throughout the construction period (Ahmed et al., 2002).
- **Involvement of different types of resources and their interactions:** Construction requires a variety of resources (Belout and Gauvreau, 2004; Tabish and Jha, 2011). A field crew member usually needs to collaborate with other crew members from another trade. Often, different trades have different priorities according to their own work scope and schedule. With a need to collaborate with each other, different tasks

could have schedule conflicts. As a result, it is common to see the occurrence of communication issues among different trades on construction jobsites.

4.2 Features of field context

In accordance with the listed characteristics of construction work, context for field crews should have the following features.

- **Context should be concise:** A large amount of information is transmitted on construction jobsites, whereas not all the information is of interest for field crews. To avoid overwhelming field crews with too much information, it is important to make the provided context concise.
- **Context can change over time:** Context can be static or dynamic. Static context information (e.g., a person's date of birth) remains the same, whereas dynamic context information (e.g., a person's location) could change over time. Context information should reflect the real state of the situation in order to be valid at a certain point in time.
- **Context has different representation formats:** Context information can be represented in various formats. For example, a field crew member's location can be represented using geographical coordinates or using a location based on a building floor plan. Another example is that a building component can have its representation in a two-dimensional drawing or in a three-dimensional model. Therefore, it is necessary to differentiate and integrate the different formats of context information with the same meaning.
- **Context is imperfect:** Context is also characterized by imperfection in that context information may be error-prone or inconsistent. For instance, a wrong supporting frame is given to field crews for the installation of a precast wall.
- **Context can be explicit or implicit:** Context is explicit when it can be directly defined (e.g., a crew member's age) or sensed (e.g., temperature). Context is implicit when it needs to be derived from explicit context; for example, a task's status is implicit because it needs to be discerned based on other context, such as

weather. For instance, if there is a heavy rain, a field task is likely to be halted. Explicit context is also referred to as low-level context in this study, while implicit context is referred to as high-level context.

5. Context for construction field crews

Considering the features of field context and the characteristics of construction jobsites, we adjusted previous definitions and suggest a definition of field context for the construction environment. Further, applying the adjusted definition, we categorized field context into an initial list of 17 different context types to form a taxonomy.

5.1 Definition of the field context

The adjusted definition of field context is as follows:

> *Context information (for construction crews) is up-to-date information about the situation of construction field crews that could influence the process or outcome of field tasks within a certain time frame.*

This adjusted definition is different from existing definitions by emphasizing two aspects.

- **Timeliness:** The identified context should reflect the latest updates. It may also help to have a prediction of the state of the situation beyond the current time point. For example, when a crew is patching a hole on a drywall, if they are not aware that a piece of equipment will be installed inside the enclosed space, they may need to chop a hole afterwards.
- **Relevancy:** Any information that conforms with this definition should be considered as field context. When the identified context information is communicated with field crews, they are supposed to be explicitly informed of what task to do and how they should perform the task. When the context information is up-to-date and correct, it should be helpful for getting the building components installed onto the final building product in a value-adding manner.

5.2 Construction context taxonomy

Since this study applies conceptual categorization, we identified specific context types and built a construction context taxonomy following the adjusted definition. In addition to the preliminary list of context types we identified through the preliminary study, we referred to existing construction categorization schemes (i.e., the POP, AROW and AFM model) and prior studies on what information is of interest to construction field crews to form an initial context taxonomy. Figure 3 shows the overall structure of the context taxonomy.

The first layer is the field context by itself.

On the second layer, the field context is subdivided into four categories: product, organization, process, and external factors. The product, organization and process categories correspond to the POP model, which describes that a physical product is delivered by some field crew through some process. Moreover, the external factors context consists of information that is not captured by the POP models.

Moving to the third layer, these four categories are further subdivided into specific context types because field crews can relate to and collect specific context information more easily. Product context includes design, material, and component. Organization context includes crew composition and crew member. Process context includes task, schedule, and equipment and tools. External factors include technical and safety specifications, industry policy, computing entity, work zone, site condition, weather, cultural and legal constraints, public relations, and nearby objects. Now that these context types have been identified based on prior studies and preliminary field studies to form an initial list, the following section discusses the validation and presents detailed explanations of each context type.

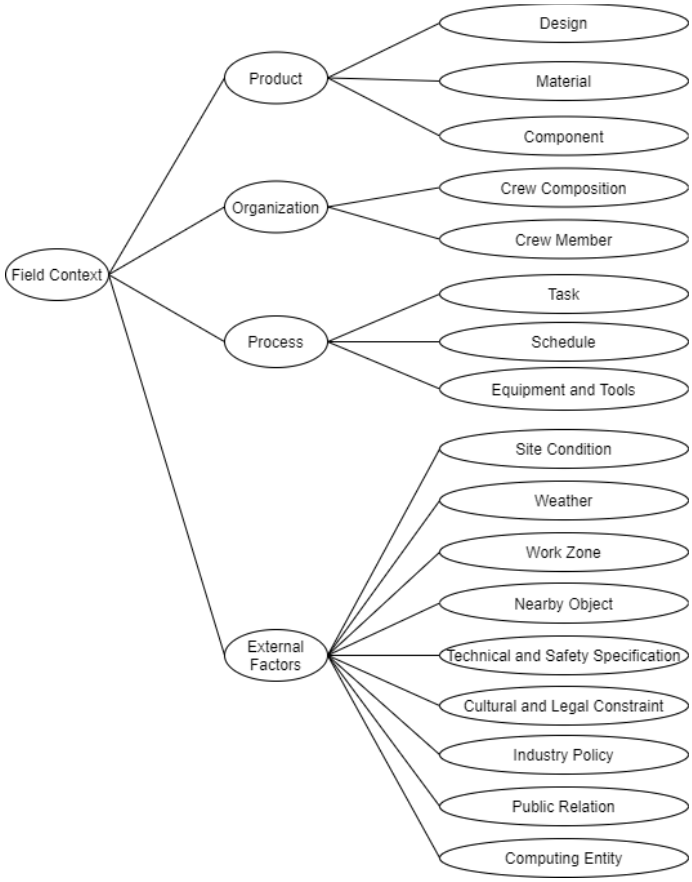

Figure 3. A context taxonomy for construction field crews

6. Methodology for validating the context taxonomy

To validate the construction context taxonomy, this study applied a qualitative approach to obtaining feedback on the 17 context types listed above by conducting semi-structured interviews with experienced industry professionals. The aim of the validation was to find out from the professionals whether the 17 context types make sense and they had experienced them, but also to determine whether the professionals could think of missing context types, i.e., situations they had experienced in the field that cannot be described by the 17 types.

6.1 Interview design

Given the restrictions on face-to-face meetings, this study conducted phone interviews with industry professionals to leverage their experience and knowledge in obtaining feedback

on the 17 context types. All the participants joined the study voluntarily. Moreover, every participant was interviewed independently in March and April, 2020.

The interviews were semi-structured. We started each interview with preset questions to ensure common feedback across all participants and, depending on the answers and areas of expertise and experience of the participant, we followed up with additional questions formulated on the fly to take advantage of each participant's unique knowledge. Questions consisted of an opening question, 17 specific questions for each context type, and an ending question.

The opening question was intended to ask about overall opinions on context from field crews. Thereafter, for each context type, there was a specific question asking about frequently seen problems with it. The reason for having a specific question for each context type was to test how frequently the participants had seen problems related to a type, since more frequent experiences would imply higher importance of a context type. At the end of each interview, we asked about whether the participant could think of any other type of information which should have been considered as context for construction field crews.

6.2 Demographic information of interviewees

Fifteen industry professionals were invited and accepted to join this study. Table 3 shows the demographic information of these interviewees. The interviewees were from five construction companies around the world: three in the United States, one in China, and one in France. Eight interviewees had worked in the construction industry for over fifteen years and the average duration in the construction industry was about twelve years. As indicated by their roles, all interviewees worked closely with construction field crews and were intimately familiar with field work.

Table 3. Demographic information of participants

Gender	
Male	13 (86.7%)
Female	2 (13.3%)
Years in Construction	
Less than 3 years	0
3 years to 5 years	3 (20%)
5 years to 15 years	4 (26.7%)
15 years to 30 years	8 (53.3%)
More than 30 years	0
Roles	
Project manager	6 (40%)
Lean practice leader	2 (13.3%)
Superintendent	4 (26.7%)
Foreman	1 (6.7%)
Project engineer	2 (13.3%)

7. Interview results

Interview responses were coded as shown in Table 5 in the Appendix section**.** For each context type, codes were made to record the reported problems, and the frequency for each code was counted.

From Table 5 in the Appendix section, it can be concluded that all the 17 context types were endorsed by the interviewees. In addition, the interviewees confirmed that the context types could influence the process or outcome of field tasks for construction field crews.

7.1 Product context

Product context provides information about what will be built by field crews. It includes intangible context (design) and tangible context (material and component). The interview results show that the reported problems were generally associated with the availability,

quality and consistency of the product context. In addition, some interviewees mentioned constructability issues.

7.1.1 Design

Design information is critical for construction field crews to ensure that the delivered product conforms with the client's requirements. Design information can be in the form of 2D drawings or 3D models. Geometric and dependency information of building objects is embedded in the design. The most reported problem was the inconsistency of design information as the design gets revised while field crews are not informed of the change in time. The following statement provide evidence for these inconsistencies: "…there are late changes of design" and "…outdated drawings are used". Besides, some interviewees reported that "…required dimensions are missing", "…dimensions are not available", and "…design does not meet specifications", which indicated that missing or erroneous design information was a common problem. Another issue mentioned was that, even though the design information was updated and correct, the design might not be constructible due to unfavorable situations on the jobsite.

7.1.2 Material

Construction field crews need to know what materials to use, whether materials are available and whether materials will require interim processing or storage at jobsites (Song et al. 1995).

The most reported problem from the interviews was the unavailability of materials. Besides, materials were reported to be unqualified; for example, one interviewee reported that "…the material does not comply with onsite testing". Moreover, materials could be wrong or might not comply with specifications.

7.1.3 Component

Construction field crews should have information (e.g., dimensional information) of the physical components that they need to install onto the final building product (Koo and Fischer, 2000). Components are different from materials in that components are delivered to the site, ready for installation without any further processing.

Interviewees reported problems including unavailable components or receiving the wrong component. Statements included "…the component does not meet specifications" and "…workers need to verify if it is the right component". Moreover, construction field crews might have difficulty in fitting a component into its designated position. The component was reported to be unqualified sometimes.

7.2 Organization context

The organization context provides information about who will perform tasks on construction jobsites (Mitropoulos and Memarian, 2012). It includes two types of context: crew composition and crew member. The problems were reported to be associated with crew members' experience, skill, behavior, and physical conditions.

7.2.1 Crew composition

Crew composition includes information about the number of crew members and the mix of skill levels and experience (Ahmadian Fard Fini et al. 2016).

Interviewees reported that the unbalanced mix of experience and skills among crew members could be a problem. Statements included that "…it is important to have the right people and right skills" and "…it helps to know where those crews come from". Some interviewees mentioned that the size of the crew could be too small or too large. Moreover, frequent mobility of crew members could have negative impacts.

7.2.2 Crew member

A field crew member is a part of context because of his/her physical presence. For example, when a crane driver is lifting materials, he/she should pay careful attention to whoever is nearby to ensure that nobody is beneath the path of the lifted material to avoid the risk of getting hit by the lifted materials. Moreover, field crew members' behavior information is important in that it helps measure productivity, evaluate work progress, and improve health and safety management (Akhavian and Behzadan 2016). A behavior can be walking, painting, idling, hammering, etc. Crew members' behavior information can be used and analyzed to improve construction productivity (Escorcia et al. 2012). In addition, Chau et al. (2004) showed that physical conditions (e.g., age, life conditions) of construction crew members have an impact on the causes and severity of occupational injuries. Gatti et al. (2014) showed a relationship between physical condition (e.g., heartbeat) and task-level worker productivity.

In summary, the interviews showed that a frequent problem for this context type was that individual crew members lacked the experience and skills for their assigned tasks. As one interviewee mentioned, "…it is important to have the right person with right skills to do the job". In addition, misbehavior can be problematic; for example, one interviewee reported that "…there were violations of PPE (personal protective equipment) compliance". Bad emotions of crew members were also mentioned as negatively affecting field tasks.

7.3 Process context

Process context provides information about how construction field crews perform tasks on construction jobsites. It includes three context types: task, schedule, and equipment and tools. For process context, similarly, problems would arise when context is missing, wrong, unqualified, or changed. Nevertheless, there are problems specific to each context type, respectively.

7.3.1 Task

Task information is a general description of what a crew member needs to accomplish for a task. Interviewees reported that sometimes the objective of the task was not clear to crew

members or the task information could be wrong. One interviewee stated, "…superintendent is not sharing the right information, and the objective is not clear to deliver certain quality of work" and "…superintendent is not communicating the right expectation".

7.3.2 Schedule

Schedule context specifies at what time a task was/is/will be executed by field crews. Construction crews should have detailed daily schedules. For example, the start and end time for a construction crew member to finish a certain task should be specified or crew members should be informed of the time for the materials to be delivered. When there is a need for collaboration between different crews, their schedules could have interdependencies. For example, a ceiling crew and a wire crew should work collaboratively to figure out the dependency between their schedules.

Moreover, schedule context reflects the temporal status (i.e., past, present, or future) of other context types. The "past" status provides a historical record of context information. The "current" status refers to the information as of now. The "future" status predicts what should take place later. It should be noted that construction tasks may be subject to changes. An understanding of the future context can be helpful in that it will allow field crews to have a sense of possible scenarios and prepare beforehand.

As the coding table (in the Appendix section) shows, conflicting schedules for different tasks was the most reported problem. One interviewee reported that "…schedules of different trades are stacking on top of each other". Another mentioned "…most of the time, schedule is not broken down into enough detail". Furthermore, last-minute changes could jeopardize the original schedule.

7.3.3 Equipment and tools

Another type of physical context is the information associated with equipment and tools (Lim and Alum 1995; Yates and Guhathakurta 1993). To accomplish certain tasks,

construction field crews need to know what equipment and tools to use, and whether the equipment and tools are available and in good condition.

For equipment and tools, it was reported that breakdowns and damage were common problems; for example, one interviewee mentioned that "… equipment and tools get lost, broken or damaged". In addition, field crews might lack necessary equipment and tools or be given wrong ones. An interviewee pointed out, "…the problems lie with the availability, maintenance, reliability and sometimes damages to the equipment and tools". Moreover, it was also reported that there is often a lack of training for construction crew members, and therefore, they do not know how to appropriately operate the equipment and tools.

7.4 External factors context

External context provides information from beyond the crew level. It includes site condition, weather, work zone, nearby objects, technical and safety specifications, cultural and legal constraints, industry policy, public relations, and computing entity. Not surprisingly, the problems reported for external factors were more diverse than for the other three main context types.

7.4.1 Site condition

Site conditions affect construction field tasks (Al-Momani 2000). Site condition context refers to the condition of the land where a construction jobsite is located. It includes the physical, geophysical, environmental, ecological, geotechnical, and archaeological conditions. Based on the interview responses, aside from the lack of access to a jobsite, there are environmental issues due to the existence of contaminated soil, underground unforeseen conditions, and bad air/lighting quality. As one interviewee put, "…site condition is one of the biggest variables". The most reported problem was the lack of access to sites.

7.4.2 Weather

Many construction tasks are performed in an outdoor environment. Therefore, they are heavily affected by weather conditions (Moselhi et al. 1997). For example, in the case of heavy rain, all outdoor tasks need to be paused. As to the weather context, interviewees reported that the occurrence of adverse weather conditions, including "high or low temperature, wind, freezing condition", could prevent construction tasks from proceeding.

7.4.3 Work zone

Work zone context provides information about the space where construction field crews perform tasks on site. More specifically, it includes information about absolute locations, relative locations, and interdependent relationships among different components (Akinci et al. 2002). The work zone context is used to determine where the finished product fits in the final building product relative to its surrounding objects.

As to the work zone, it was reported that there might be hazards for construction crew members. Moreover, misalignment of objects and lack of space could make it difficult for a task to proceed. Statements from interviewees included "…the work zone has to be safe" and "…it is important to ensure the availability of enough space for the crews".

7.4.4 Nearby object

Nearby objects could include stacking materials, equipment, tools, components, openings, etc., which are not used by a task but could prevent crew members from performing their task as planned. For instance, crew members should be aware of any openings nearby to avoid getting hurt by falling into the opening.

The most frequently reported problem was the existence of trash, hazards, unused equipment and tools, and stacked materials. For example, interviewees reported that "materials not relevant at the time", "equipment that could hinder the installation", "trash, broken system" and "overhead cables" could be nearby objects.

7.4.5 Technical and safety specification

Field crews should strictly follow technical and safety specifications to ensure that their finished work satisfies quality requirements and, more importantly, that no safety incidents will occur throughout the construction process. One interviewee emphasized the importance of following technical and safety specifications as follows, "…if the worker is not following our safety process, there is zero tolerance". Some interviewees also reported that technical and safety specifications usually do not change that often.

7.4.6 Cultural and legal constraints

Culture and laws (e.g., wage laws) can affect how construction field crews perform tasks (Khan, 2014). Most interviewees reported that they have been confronted with adverse collaborative environments. For example, one interviewee mentioned that "…language can be an issue, and work ethics may not be the same", and another interviewee reported that "…it is important to encourage work partners to work with each other". In addition, some interviewees reported that field crews should be informed of relevant contractual information.

7.4.7 Industry policy

Construction industry policies can vary in different countries, regions, or cities. When performing tasks, crew members may be constrained by certain industry policies according to the location of their project. Differences in industry policies between project locations can be confusing for construction field crews. However, industry policies usually do not change frequently. As one interview pointed out, "…industry policy does not change much, and therefore there are not many problems associated with it". However, construction crew members should still be aware of "policy updates" if there are any.

7.4.8 Public relation

Through construction processes, construction crews may need to maintain relationships with the government and the neighboring community. Interviewees reported that they have had problems with complaints from neighbors. For example, interviewees reported that

"…neighbor is not happy to have a construction site nearby" and therefore "…it is important to work on community involvement and engagement".

7.4.9 Computing entity

With the implementation of more computational devices on jobsites, computing entity becomes important for field crews. Computing entity includes network, software, hardware, etc.

Interviewees recognized the usefulness of computing entities. However, training for crews for these devices is often insufficient. As reported by one interviewee, 'it is critical to provide them (field crews) with the correct level of training'. In addition, poor integration of these computing tools makes implementation difficult. One interviewee reported that '…field crews are getting too many tools, and the tools end up taking a lot of time'.

7.5 Summary

Each of the 17 context types in the context taxonomy was endorsed by all of the interviewees. In addition, these interviewees shared their experiences on frequently seen problems with each context type. However, as shown in the coding table in the Appendix section, the interviewees did not see every context type as equally important for field crews. Specifically, every interviewee reported problems regarding the design, component, material, crew composition, crew member, task, schedule, equipment and tools, site condition, work zone, weather, nearby objects, public relation, and computing entity context. Some interviewees reported that the cultural and legal constraints, industry policy, and technical and safety specification context led to problems less frequently because they change rarely.

8. Study of field problems using identified context types

In addition to the semi-structured interviews with industry professionals, this study analyzed daily field logs obtained from a project performed by a concrete sub-contractor

located in Maryland, US as an empirical test of the context taxonomy describe above. The daily logs were recorded from May 18, 2020 to September 10, 2020. The work scope of the concrete crew was the concrete foundation of the project. From these daily logs, a total of 112 problem instances for the concrete crew were reported. The researchers mapped the reported problems to the 17 context types. It was found that every problem could be successfully categorized. For example, it was reported that "rebar GR100 was not delivered on time", which was categorized as a material-related problem; the problem "crew members cannot work in Pour #2 since plumbers and electricians are working in Pour #2 area" was related to "work zone". Table 4 lists the context types associated with the 112 problems and the number of occurrences for each context type.

As can be seen from the table, the reported problems were associated with "equipment and tools", "weather", "task", "material", "schedule", "work zone", "crew composition", "site condition", "design" and "component". While all reported problems could be categorized with the identified context types, due to the specific scope of the construction work in the case study and the limited number of problems being reported from one project, not all context types were present on this project.

Furthermore, among the context types, 'equipment and tools' (23, 21%), 'task' (18, 16%), 'schedule' (17, 16%) and 'work zone' (24, 22%) had the most reported problems. The percentages are shown in Figure 4. The variation in the number of occurrences for different context types indicates that, for a certain project, different context types may be of varied significance. Hence, construction field crews may benefit from being aware of the differences when preparing for daily tasks.

Table 4. Mapping of field problems to context types

Context Type for Reported Problem	Occurrences of Reported Problems
Equipment and tools	23
Weather	2
Task	18
Material	7
Schedule	17
Work zone	24
Crew composition	7
Site condition	6
Design	4
Product	2
Component	2
Total occurrences of mapped context types	112

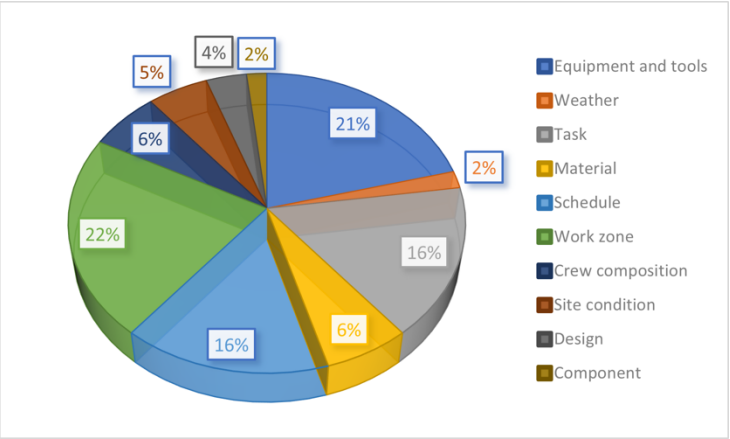

Figure 4. Percentage of occurrence of each context type

9. Discussion

Previous studies have pointed out that poor communication could negatively impact the schedule performance of construction projects (Hwang and Yang, 2014; Habibi et al., 2018). This study pays special attention to communication with construction field crews whose work directly adds value onto the final building product. This paper discussed the meaning of context and presents a context taxonomy for construction field crews. In the discussion section, we reflect on the process of building the context taxonomy and the method to validate it.

Since context is the core term in this study, we found it necessary to clarify its meaning before moving any further. Context originated from the domain of computer science (Dey, 2001; Hull et al., 1997; Chen, 2002), and was later studied by researchers in other fields (e.g., healthcare) with the intention to develop context-aware applications. These attempts to implement context in a different field require consideration of unique characteristics of that field. To clarify the definition of context for field crews, therefore, this study adjusted previous definitions by analyzing the characteristics of construction field work. Although the adjusted definition could help discern whether a type of information should be considered as field context, the challenge still existed as we attempted to build a context taxonomy.

Moreover, following the differentiation of two types of categorization schemes (i.e., operational categorization and conceptual categorization) proposed by Van Bunningen et al. (2005), this study was aimed at providing a conceptual categorization of context for field crews. "Categorization/Classification scheme" and "taxonomy" are often used interchangeably in the literature (Oberländer et al., 2019), and this paper does not rigorously differentiate these two terms. We use the term "taxonomy" to represent conceptually and/or empirically derived types of objects in a broad sense. By developing a context taxonomy, this study standardizes context types to provide consistency among users of context information (Gregor, 2006).

As to the development of the context taxonomy, this study implemented an empirical-to-conceptual process to identify context types. This paper generalizes the development process of the taxonomy and summarizes four major steps as below. This taxonomy development process might help future research to develop a context taxonomy for a different entity in the construction domain.

1. Set the goal for developing the context taxonomy. In this study, the goal was to provide a common understanding of context for construction field crews. With a context taxonomy, communication of context information could be consistent.
2. Conduct field studies to acquire necessary data. We visited four construction sites from which 40 instances of field task problems were observed. For each visit, we

recorded problems where field crews were not informed of context information and its changes in a timely manner. Categorization of these 40 problems led us to our preliminary list of 11 context types: design, weather, material, work zone, equipment and tools, component, crew member, crew composition, public relation, technical and safety specification and schedule. These 40 problems were categorized into some common types, which implies that there might be a limited number of context types. However, at this step, we were not sure about what the limited number is.

3. Complement the preliminary list with findings from the literature. Coupled with the empirical evidence in step 2, we referred to the literature to deduce other context types (if any); for example, existing construction models (i.e., the POP, AROW and AFM models) were reviewed. Through this deductive process, we were able to expand the preliminary list of context types to 17 context types: design, material, component, crew composition, crew member, task, schedule, equipment and tools, technical and safety specifications, industry policy, computing entity, work zone, site condition, weather, cultural and legal constraints, public relations, and nearby objects. Thereafter, we reviewed at least 5 papers regarding each context type to further understand how it would affect field crews when performing tasks.
4. Validate the context taxonomy through both qualitative and quantitative assessment. This study deployed a qualitative method by performing semi-structured interviews coupled with a quantitative analysis based on data extracted from daily logs of one project. The feedback on each listed context type from the interviewees helped us evaluate whether it is reasonable to include the context type in the taxonomy. It was critical that we asked every interviewee whether they could think of a different context type that had not been included in the taxonomy. The answer to this question might add another context type to the taxonomy.

10. Conclusions

Field crews can benefit from being informed of context information and its changes in a timely manner (Jesus and Howitt, 1998), but the lack of a shared understanding of field

context makes it difficult to do so. This study presents an adjusted definition of field context for construction field crews and a context taxonomy consisting of 17 specific context types. Semi-structured interviews with 15 experienced industry professionals from sophisticated construction companies and an empirical study on an ongoing project validated these context types. The interviews collected frequently seen problems with every context type to determine how each context might influence the outcome or the process of field tasks. Every context type was endorsed by the interviewees to be of interest for field crews. In addition, the interviewees did not report any other context types that are not listed in the 17 context types. The empirical study showed that every reported problem from the ongoing project could be mapped to one of the 17 context types.

With the adjusted definition of context and a context taxonomy, this study could provide consistency for field crews when communicating context information. By building the common ground, this study facilitates future research on how to utilize context and improve context-awareness for construction field crews. As for practical implementation, construction field crews can refer to the context taxonomy to figure out what information should be considered and communicated to accomplish their tasks. For example, field work instructions could be developed based on the context information so that context-awareness of field crews could be enhanced.

This study has the following limitations and future work. First, the semi-structured interviews were done with only 15 industry professionals in the building sector. Although the interviewees are from different construction companies and have an average of more than 12 years of work experience, the qualitative analysis could still be doubted due to the small number of interviewees. Hence, the validation should be extended to more interviewees. Second, the quantitative assessment in section 8 has its limitation since we only made analysis based on partial data from one project. The total number of reported problems from the project was not enough to reflect each of the 17 context types. Third, this study did not differentiate context types for different types of field work. The importance of each context type may vary for concrete crews and façade crews; meanwhile, the importance of different context type for the same type of field work could also vary.

Hence, future work should understand how each context type affects the tasks for different crews. An understanding of the effects could be leveraged to enhance production control. For instance, it helps to understand how crew members' behavior information affects their productivity. Moreover, construction field crews could benefit from an increased awareness of context when performing tasks. We will explore how to develop field work instructions that could effectively communicate context and its changes to construction field crews.

Acknowledgement

This study was funded by the Seed Funding provided by the Center for Integrated Facility Engineering (CIFE) at Stanford University. Thanks to Bouygues Construction, Accu-Crete, DPR Construction, Ruldoph & Sletten, we were able to conduct field studies on multiple projects. In addition, we thank the 15 industry professionals who participated in the interviews and provided insightful answers to the interview questions.

Appendix

Table 5. Results of coding for the interview responses

	Context types	Codes	Frequency
Specific questions	Design	Lack design information	3
		Errors in design	4
		Inconsistency between design iterations	10
		Design cannot be constructed	4
	Component	Component not available	7
		Wrong component	6
		Fitting problems (Constructability)	7
		Unqualified component	2
	Material	Material not available	7
		Wrong material	3
		Unqualified material	7
		Lack material specification information	1
	Crew composition	Inappropriate size of crew	2
		Unbalanced experience and skills	5
		Diverse backgrounds	3
		Mobility of crew members	1
	Crew member	Lack of experience and skills	2
		Crew members' misbehaviors	5
		Crew members' bad emotions	2
	Task	Task objective is not clear	5
		Wrong task information	2
		Too much/little information	2
	Schedule	Last-minute change of schedule	2
		Schedule is not detailed enough	1
		Conflict of schedules	5
	Equipment and tools	Breakdown/damage of equipment and tools	7
		Lack of equipment and tools	3
		Unqualified equipment and tools	2
		Wrong equipment and tools	2
		Lack access	2
	Site condition	Lack access	5

		Unfavorable soil and underground condition	4
		Bad air quality	1
		Bad lighting quality	1
	Work zone	Space is not enough	3
		Existence of hazards	5
		Work zone is not tidy or clean	2
		Misalignment	2
	Weather	Adverse weather (e.g., rain, wind)	15
	Irrelevant nearby object	Existence of trash	3
		Existence of hazards	3
		Equipment/tools nearby but not used	1
		Material nearby but not used	1
	Culture and law	Violation of contractual information	3
		Adverse collaborative environment	10
	Industry policy	Inconsistency of industry policy	3
	Technical and safety specification	Violation of tech and safety specs	4
		Inconsistency of tech and safety specs	3
	Public relations	Complaints from neighborhood	14
	Computing entity	Connectivity issue	2
		Poor integration of tools	3
		Lack training	3
Ending question	Another context?	Almost everything	15

References

Abowd, G., Dey, A., Brown, P., Davies, N., Smith, M., and Steggles, P. (1999). "Towards a better understanding of context and context-awareness." *International symposium on handheld and ubiquitous computing*, Springer, Berlin, Heidelberg, 304–307.

Abowd, G. and Mynatt, E. (2000). "Charting past, present, and future research in ubiquitous computing." *ACM Transactions on Computer-Human Interaction (TOCHI)*, *7*(1), 29–58.

Adomavicius, G. and Tuzhilin, A. (2011). "Context-aware recommender systems." *Recommender systems handbook*.

Ahmadian Fard Fini, A., Rashidi, T.H., A. A., and Travis Waller, S. (2016). "Incorporating multi-skilling and learning in the optimization of crew composition." *Journal of construction engineering and management*, *142*(5), 04015106.

Ahmed, S.M., Azhar, S., Castillo, M. and Kappagantula, P., 2002. Construction delays in Florida: An empirical study. *Final report. Department of Community Affairs, Florida, US.*

Akhavian, R. and Behzadan, A. (2016). "Smartphone-based construction workers' activity recognition and classification." *Automation in Construction*, 71.

Akinci, B., Fischer, M., and Kunz, J. (2002). "Automated generation of work spaces required by construction activities." *Journal of construction engineering and management*, *128*(4), 306–315.

Al-Momani, A. (2000). "Construction delay: a quantitative analysis." *International journal of project management*, *18*(1), 51–59.

Arbulu, R., Tommelein, I., Walsh, K., and Hershauer, J. (2003). “Value stream analysis of a re-engineered construction supply chain.” *Building Research Information*, *31*(2), 161–171.

Aziz, Z., Anumba, C.J., Ruikar, D., Carrillo, P. and Bouchlaghem, D., 2006. Intelligent wireless web services for construction—A review of the enabling technologies. *Automation in Construction*, *15*(2), pp.113-123.

Belout, A. and Gauvreau, C., 2004. Factors influencing project success: the impact of human resource management. *International journal of project management*, *22*(1), pp.1-11.

Bettini, C., Brdiczka, O., Henricksen, K., Indulska, J., Nicklas, D., Ranganathan, A., and Riboni, D. (2010). “A survey of context modeling and reasoning techniques.” *Pervasive and Mobile Computing*, *6*(2), 161–180.

Brown, P.J., 1995. The stick-e document: a framework for creating context-aware applications. *Electronic Publishing-Chichester-*, 8, pp.259-272.

Brown, P., Bovey, J., and Chen, X. (1997). “Context-aware applications: from the laboratory to the marketplace.” *IEEE personal communications*, *4*(5), 58–64.

Chau, N., Gauchard, G., Siegfried, C., Benamghar, L., Dangelzer, J., Français, M., Jacquin, R., Sourdot, A., Perrin, P., and Mur, J. (2004). “The construction industry as a loosely coupled system: implications for productivity and innovation.” *Construction Management Economics*, *77*(1), 60–66.

Chen, Harry. "An intelligent broker architecture for context-aware systems." *UMBC Student Collection* (2002).

Chen, Y. and Kamara, J.M., 2011. A framework for using mobile computing for information management on construction sites. *Automation in construction*, *20*(7), pp.776-788.

Dey, A. (2001). “Understanding and using context.” *Personal and ubiquitous computing*, *5*(1), 4–7.

Dubois, A. and Gadde, L. (2002). “The construction industry as a loosely coupled system: implications for productivity and innovation.” *Construction Management Economics*, *20*(7), 621–631.

Escorcia, V., Dávila, M., Golparvar-Fard, M., and Niebles, J. (2012). “Automated vision-based recognition of construction worker actions for building interior construction operations using rgbd cameras.” *Construction Research Congress 2012: Construction Challenges in a Flat World*.

Garcia-Lopez, N. and Fischer, M. (2016). “A construction workflow model for analyzing the impact of in-project variability.” *Construction Research Congress*.

Gatti, U., Migliaccio, G., Bogus, S., and Schneider, S. (2014). “An exploratory study of the relationship between construction workforce physical strain and task level productivity.” *Construction management and economics*, *32*(6), 548–564.

Gidado, K. (1996). “Project complexity: The focal point of construction production planning.” *Construction Management Economics*, *14*(3), 213–225.

Gray, P. and Salber, D. (2001). “Modelling and using sensed context information in the design of interactive applications.” *IFIP International Conference on Engineering for Human-Computer Interaction*, Springer, Berlin, Heidelberg, 317–335.

Gray, P. and Salber, D. (2004). “Middlewhere: a middleware for location awareness in ubiquitous computing applications.” *ACM/IFIP/USENIX International Conference on Distributed Systems Platforms and Open Distributed Processing*, Springer, Berlin, Heidelberg, 397–416.

Gregor, S., 2006. The nature of theory in information systems. *MIS quarterly*, pp.611-642.

Gross, T. and Specht, M. (2001). “Awareness in context-aware information systems.” *Mensch Computer 2001*, Vieweg+ Teubner Verlag, 173–182.

Guo, S.J., 2002. Identification and resolution of work space conflicts in building construction. *Journal of construction engineering and management*, *128*(4), pp.287-295.

Habibi, M., Kermanshachi, S. and Safapour, E., 2018, April. Engineering, procurement and construction cost and schedule performance leading indicators: state-of-the-art review. *In Proceedings of Construction Research Congress* (pp. 2-4). New Orleans, Louisiana: ASCE.

Henricksen, K. (2003). “A framework for context-aware pervasive computing applications.” *In Workshop on human computer interaction with mobile devices*.

Henricksen, K., Indulska, J., and Rakotonirainy, A. (2002). “Modeling context information in pervasive computing systems.” *International Conference on Pervasive Computing*, Springer, Berlin, Heidelberg, 167–180.

Henricksen, K. and Indulska, J., 2004, March. Modelling and using imperfect context information. In *IEEE Annual Conference on Pervasive Computing and Communications Workshops, 2004. Proceedings of the Second* (pp. 33-37). IEEE.

Hull, R., Neaves, P. and Bedford-Roberts, J., 1997, October. Towards situated computing. In *Digest of papers. first international symposium on wearable computers* (pp. 146-153). IEEE.

Hwang, B.G. and Yang, S., 2014. Rework and schedule performance: A profile of incidence, impact, causes and solutions. Engineering, Construction and Architectural Management.

Jesus, M. and Howitt, I., 1998. Wireless communication and computing at the construction jobsite. *Automation in Construction*, *7*(4), pp.327-347.

Khan, A. (2014). “Effects of cultural assimilation on the performance of a construction project–evidence from uae.” *Benchmarking: an international journal*.

Koo, B. and Fischer, M. (2000). “Feasibility study of 4d cad in commercial construction.” *Journal of construction engineering and management*, *126*(4), 251–260.

Kunz, J. and Fischer, M. (2012). “Virtual design and construction: themes, case studies and implementation suggestions.

Lee, J., Song, J., Oh, K., and Gu, N. (2013). “Information lifecycle management with rfid for material control on construction sites.” *Advanced engineering informatics*, *27*(1), 108–119.

Lee, S. and Mohsen, O. (2008). “Application of context-aware computing to construction field work.” *Proceedings of the CSCE 2008 Annual Conference*, Vol. 1, 483–490.

Lei, H., Sow, D., Davis, J., Banavar, G., and Ebling, M. (2002). “The design and applications of a context service.” *ACM SIGMOBILE Mobile Computing and Communications Review*, *6*(4), 45–55.

Lim, E. and Alum, J. (1995). “Construction productivity: issues encountered by contractors in singapore.” *International journal of project management*, *13*(1), 51–58.

Mak, S., 2001. A model of information management for construction using information technology. *Automation in construction*, *10*(2), pp.257-263.

Mitropoulos, P. and Memarian, B., 2012. Team processes and safety of workers: Cognitive, affective, and behavioral processes of construction crews. *Journal of Construction Engineering and Management*, *138*(10), pp.1181-1191.

Moselhi, O., Gong, D., and El-Rayes, K. (1997). “Estimating weather impact on the duration of construction activities.” *Canadian Journal of Civil Engineering*, *24*(3), 359–366.

Mourgues, C. and Fischer, M. (2008). “A work instruction template for cast-in-place concrete construction laborers.

Oberländer, A.M., Lösser, B. and Rau, D., 2019. Taxonomy research in information systems: a systematic assessment.

Park, J. and Cho, Y.K., 2017. Development and evaluation of a probabilistic local search algorithm for complex dynamic indoor construction sites. *Journal of Computing in Civil Engineering*, *31*(4), p.04017015.

Pascoe, J. (1998). “Adding generic contextual capabilities to wearable computers.” *Wearable Computers*, IEEE, Digest of Papers. Second International Symposium, 92–99.

Perera, C., Zaslavsky, A., Christen, P., and Georgakopoulos, D. (2013). “Context aware computing for the internet of things: A survey.” *IEEE communications surveys tutorials*, *16*(1), 414–454.

Rebolj, D., Babič, N.Č., Magdič, A., Podbreznik, P. and Pšunder, M., 2008. Automated construction activity monitoring system. *Advanced engineering informatics*, *22*(4), pp.493-503.

Rosenberger, P., Gerhard, D., and Rosenberger, P. (2005). “Confronting context effects in intelligence analysis: How can mathematics help.

Rosenberger, P., Gerhard, D., and Rosenberger, P. (2018). “Context-aware system analysis: Intro- duction of a process model for industrial applications.” *ICEIS*, Vol. 2, 368–375.

Ryan, N., Pascoe, J., and Morse, D. (1998). “Enhanced reality fieldwork: the context-aware archaeological assistant.” *Computer applications in archaeology*, Tempus Reparatum.

Schilit, B., Adams, N., and Want, R. (1994). “Context-aware computing applications.” *1994 First Workshop on Mobile Computing Systems and Applications*, IEEE, 85–90.

Schilit, Bill N., David M. Hilbert, and Jonathan Trevor. "Context-aware communication." *IEEE Wireless Communications* 9, no. 5 (2002): 46-54.

Singhvi, V. and Terk, M. (2011). “Prophet: a contextual information system framework.” *CIB REPORT*, Vol. 284, 318.

Snidaro, L., García, J., and Llinas, J. (2015). “Context-based information fusion: a survey and discussion.” *Information Fusion*, 25, 16–31.

Song, J., Haas, C., and Caldas, C. (1995). “Tracking the location of materials on construction job sites.” *Journal of construction engineering and management*, *132*(9), 911–918.

Tabish, S.Z.S. and Jha, K.N., 2011. Identification and evaluation of success factors for public construction projects. *Construction Management and Economics*, *29*(8), pp.809-823.

Thomas, H. and Napolitan, C. (1995). “Quantitative effects of construction changes on labor productivity.” *Journal of construction engineering and management*, *121*(3), 290–296.

Truong, B., Lee, Y., and Lee, S. (2005). “Modeling and reasoning about uncertainty in context- aware systems.” *IEEE International Conference on e-Business Engineering (ICEBE’05)*, IEEE, 102–109.

Van Bunningen, A.H., Feng, L. and Apers, P.M., 2005, April. Context for ubiquitous data management. In *International Workshop on Ubiquitous Data Management* (pp. 17-24). IEEE.

Wang, H., Boukamp, F., and Elghamrawy, T. (2011). "Ontology-based approach to context representation and reasoning for managing context-sensitive construction information." *Journal of Computing in Civil Engineering*, *25*(5), 331–346.

Ward, A., Jones, A., and Hopper, A. (1997). "A new location technique for the active office." *IEEE Personal communications*, *4*(5), 42–47.

Winograd, T. (2000). "Architectures for context." *Human-Computer Interaction*, *16*(2), 401–419.

Yates, J. and Guhathakurta, S. (1993). "International labor productivity." *Cost Engineering*, *35*(1), p.15.

Zimmermann, A., Lorenz, A., and Oppermann, R. (2007). "An operational definition of context." *International and Interdisciplinary Conference on Modeling and Using Context*, Springer, Berlin, Heidelberg, 558–571.

Chapter 3: Development of A Construction Context Information (CCI) Model for Construction Field Crews

Abstract: Mobile computing is becoming increasingly popular on construction jobsites to improve communication for field crews. Hence, great attention has been paid towards the development and implementation of mobile computing technologies on construction jobsites. In the dynamic construction jobsite environment, mobile computing technologies would be most beneficial if the system was aware of and adapted to the field context and its changes. However, there is no model built upon validated context types that could serve as the basis of a context-aware mobile communication system for field crews. This paper introduces a construction context information (CCI) model to serve as a formal representation of attributes and relationships of the field context to facilitate field communication by enhancing context-awareness on construction jobsites. The CCI model is represented in the form of OWL (Web Ontology Language). The CCI model is built upon 17 validated context types and defines their associated attributes and relationships. Moreover, this paper shows how the CCI model supports the generation of general reasoning rules to enable inferences of implicit high-level context information (e.g., the status of a task) from explicit low-level context information (e.g., weather). Overall, the CCI model could be used to develop a context-aware system that helps observe, interpret, reason about, and communicate context information for construction field crews. Future work should explore how context-awareness could be enhanced through the use of the CCI model and to what extent field crews could benefit from the enhancement.

Key words: Context; Construction; Field Work; Field Crew; Ontology

1. Introduction

Recent years have witnessed a growing body of research on the implementation of Information and Communication Technology (ICT) in the construction industry (Bowden et al., 2006; Adriaanse et al., 2010; Lam et al., 2010). ICT has been utilized to enhance the efficiency of communication in the domain of construction information management and

there have been attempts to extend its implementation to jobsites to assist field crews in performing daily tasks (Kasim and Ern, 2010; Perkinson et al., 2010; Kasim, 2011). Regardless of those technological advancements (Acar et al., 2005; Aziz et al., 2006; Anumba et al., 2008), field crews still lack access to the right information at the right time (Fathi et al., 2012). Thanks to the emergence of more affordable mobile devices and increasingly reliable wireless networks, mobile computing has become increasingly promising to enhance communication for field crews (Arruda et al., 2014; Sattineni and Schmidt, 2015). Since construction field tasks are information-intensive and dynamic (Anumba et al., 2008), communication using mobile computing technologies is inherently difficult for field crews without being aware of and adapting to the field context and its changes. For this study, we define context information as "up-to-date information about the situation of construction field crews that could influence the process or outcome of field tasks within a certain time frame".

Many studies have explored context-aware systems to facilitate communication on construction jobsites (Chen and Kotz, 2000). Specifically, Bae et al. (2013) developed a context-aware system to provide vital project information, such as the location of elements and the expected quality of building elements, to on-site users. Omar and Ballal (2009) developed a system to support construction project supply chain logistics by integrating context information such as user context, environmental context, and project context. These systems showed that increased context awareness improves field communication; however, their focus was on some facets of field context (e.g., location). In a prior study, we identified 17 basic field context types and validated them through semi-structured interviews with experienced industry professionals and the categorization of 112 problems observed on a jobsite over five months. These 17 context types provide a holistic taxonomy that could be a basis for utilizing field context and develop context-aware systems for field crews. According to Yuan and Herbert (2014), a context-aware system should help observe, interpret, and reason about context information. Interpretation and reasoning provide deeper insights from observed context information. Our context taxonomy helps field crews observe and discern types of context information; however, it is not enough for interpretation and reasoning. Similar to a language requiring grammar and part of speech

to connect individual words, the validated taxonomy requires the identification of relationships and attributes of context types to have more expressiveness. For example, for the material context, it is worthwhile to know its availability (i.e., "availability" is a attribute of the material context). In addition, the material context should be related to the task context to indicate that a task consumes materials. Meanwhile, the material context should be related to the design context to indicate that physical objects align with design. Based on the defined attributes and relationships, users of context information would be able to reason about the status of the task. By the status of a task, we refer to "delayed", "completed", etc. Like Google Maps tells the user whether there is a delay and calculates how much time the delay will likely take, reasoning about context information should perform inference about whether a task is delayed and how long the delay is. Still, in the example, the status of a task would be inferred to be "delayed" if the materials are not available. Moreover, conflicts of context can also be derived; for example, a conflict arises when a material is not the same as specified in design. Therefore, to further enable interpretation and reasoning of the field context, it is necessary to specify attributes and relationships associated with these identified context types.

In this paper, we propose a Construction Context Information (CCI) model based upon the 17 context types. The model consists of attributes of and relationships between these context types. The model is in the form of an ontology encoded with OWL (Web Ontology Language). Through the creation of such an ontology-based model, this paper enhances the expressiveness of the context taxonomy so that field communication can benefit from the ability to interpret and reason about context information. Specifically, for reasoning, this paper shows that general rules can be defined using the Semantic Web Rule Language (SWRL), and, therefore, the context model supports inference of high-level context from low-level context and reasoning about conflicts of context information. In this paper, we use low-level context to refer to context that can be directly sensed (e.g., weather) or defined (e.g., availability of material) and high-level context (e.g., the status of a task) to refer to context that is implicit and requires assessment based on low-level context.

The organization of this paper is the following: section 2 presents related work to this research, including methods for modeling and reasoning about context; section 3 explains the 17 basic context types upon which the CCI model was built; section 4 details the CCI model, including the structure of the model, attributes of each context type and their relationships; then in section 5, reasoning rules are presented to show how high-level context can be derived from low-level context and to illustrate how conflicts of context information can be detected. Finally, we draw conclusions and discuss future work in section 6 followed by acknowledgements in section 7.

2. Related work

Since this paper intends to build a CCI model that supports observation, interpretation, reasoning about, and communication of context information, we reviewed the available modeling methods. The past decade has witnessed the emergence and implementation of a few context modeling methods (Strang and Linnhoff-Popien, 2004). This section examines and compares these context modeling methods. Specifically, for reasoning, we reviewed relevant studies to figure out how reasoning rules should be made to enable inference of context information.

2.1 Context Modeling Methods

To build our so-called CCI model, we had to first decide what context modeling method to use. The selection of a context modeling method should consider the requirements of building a context model and characteristics of existing modeling methods. According to Reichle et al. (2008), our CCI model should be characterized with the following features.

- Simple to implement: The CCI model should not require much effort to build.
- Machine-interpretable: The CCI model should provide machine-interpretable representations of field context. A typical approach is the use of semantic annotations to represent field context.
- Dealing with large amounts of information: Considering a construction project may involve thousands of field tasks or even more, and there are 17 context types along

with their associated attributes and relationships, the CCI model should be able to process large amounts of information.

- Reusability: The CCI model should serve as a common understanding of the field context, and these types of context information along with their attributes and relationships should be reusable.
- Extensibility: The CCI model should be extensible to include possibly new relationships and attributes of context.
- Support of reasoning: The model needs to support reasoning through which conflicts of context and high-level context could be derived.

To search for a method that satisfies the above-discussed requirements, we reviewed existing methods for modeling context information. This section compares the strengths and weaknesses of these modeling methods in representing context information.

The selection of a modeling method should consider characteristics of available methods in accordance with the above-mentioned requirements for ontology modeling. Chen and Kotz (2000), Strang and Linnhoff-Popien (2004), and Perera et al. (2013) compared different context modeling methods including key-value modeling, mark-up scheme modeling, graphical modeling, object-based modeling, logic-based modeling, and ontology-based modeling. Below we discuss the strengths and weaknesses of these context modeling methods.

- Key Value Models (KVM): The KVM method is the simplest to represent context information and good at managing a small amount of data. However, it is neither suitable for managing large amounts of data nor scalable for complex data structures.
- Markup Scheme Models (MSM): The MSM method allows efficient data retrieval and range checking, but it does not provide advanced expressive capabilities. Moreover, re-usability over different markup schemes is difficult.

- Graphical Models (GM): This method is suitable for describing the structure of context information and can hold large amounts of data. However, this method may suffer from ambiguity and may lead to misinterpretation of context.
- Logic-Based Models (LBM): The LBM method has more expressive richness than the above-mentioned methods and supports reasoning about context information. However, this method limits reusability.
- Object-Oriented Models (OOM): The OOM method improves reusability but it does not appropriately address incompleteness and ambiguity.
- Ontology-Based Models (OBM): The previous modeling methods (i.e., KVM, MSM, GM, LBM and OOM) can be considered as precursors to OBM and ontology models, which incorporate good aspects from these other modeling methods. Ontology-Based Models are suitable for representing context information in a machine-interpretable form.

Since the construction field environment engages different participants and field communication involves a high volume of information, the ontology-based modeling method is preferable for this study because it aligns with the modeling requirements to the largest extent. Modeling context in the form of an ontology makes sure that the meaning of the context information is unambiguous and, therefore, ontology-based modeling has better reasoning capabilities for information inference (Ye et al., 2007). A widely accepted definition of ontology was given by Studer et al. (1998). They defined ontology as "a formal, explicit specification of a shared conceptualization. A conceptualization refers to an abstract model of some phenomenon in the world by having identified the relevant concepts of that phenomenon". To build a context model using the ontology-based modeling method, several popular semantic web ontology languages can be used: RDF (Resource Description Framework), RDFS (Resource Description 65 Framework Schema), and OWL (Web Ontology Language) (Brickley et al., 2014). OWL is an improved version of RDFS with more expressiveness, higher machine interoperability, and richer semantics (Nurmi and Floréen, 2004). OWL can be used to explicitly represent the context types and their relationships in the context model (McGuinness and Van Harmelen, 2004). Therefore, OWL was selected in this research to build a construction context information (CCI) model.

Below, we show explanations for the key concepts that will be used later on in this paper (McGuinness and Van Harmelen, 2004).

- Class: A class defines a group of individuals that share some common properties.
- rdfs:subClassOf: A class may be a subclass of another class.
- owl:ObjectProperty: ObjectProperty can be used to state relationships between classes.
- owl:DatatypeProperty: DatatypeProperty can be used to state relationships between classes and data values.
- Individual: Individuals are instances of classes.

2.2 Context Reasoning

Context reasoning is a process in which new and relevant information to the use of applications(s) and user(s) can be deduced from various sources of context data (Bikakis et al., 2007). They classify context into high-level context and low-level context. High-level context is usually implicit and cannot be acquired directly, whereas the latter can be directly obtained from a context provider (e.g., a sensor). Low-level context can be sensed from various types of sensors (e.g., weather) or defined by users (e.g., a crew member's name). Through context reasoning, low-level context can be transformed into meaningful high-level context (e.g., the status of a task). For example, if a context-aware system detects that it is going to rain, field crews should then be aware that the task will be halted. However, sensed or user-defined context data is characterized by imperfection and uncertainty (Bettini et al., 2010). Imperfect context can be unknown, ambiguous, imprecise, and erroneous. Through context reasoning, possible inconsistencies or errors in the context information can be detected (Wang et al., 2018). For example, a field engineer finds an error in the shop drawing during the weekly planning process, which field crews should be informed of and have the error fixed before they start their task. Regarding inconsistencies of context information, for example, the context reasoning process should detect cases when the material context does not align with the design context.

3. Context types for construction field crews

The proposed CCI model was built upon our previous work that identified and validated 17 basic field context types: “design”, “material”, “component”, “crew composition”, “crew member”, “task”, “schedule”, “equipment and tools”, “weather”, “work zone”, “site condition”, “nearby objects”, “technical and safety specifications”, “cultural and legal constraints”, “industry policy”, “public relations”, and “computing entity”. These context types were categorized into four types: product, organization, process, and external factors. This section briefly describes these 17 context types.

3.1 Product context

Product context provides information about what construction field crews need to build. It can be intangible context (i.e., “design”) or tangible context (i.e., “material” and “component”). Intangible context refers to virtual product information while tangible context refers to physical product information.

1) Design context is usually in the form of 2D drawings or 3D models. It consists of geometric, dependency, and connection information of building products.
2) Material context specifies what materials to use, whether the materials are available or meet specifications, and whether the materials will require interim processing or storage at jobsites. For example, cement is an instance of material context.
3) Component context is different from material context in that a component can be installed without further processing on jobsites. For example, a precast stair is an instance of component context.

3.2 Organization context

Organization context provides information about who performs tasks on construction jobsites. It includes two types of context: “crew composition” and “crew member”.

4) Crew composition refers to the composition of a crew. It contains information about the number of crew members and the mixture of their skill levels and experience.
5) A field crew member is a part of context because of his/her physical presence, actions, and physical conditions.

3.3 Process context

Process context provides information about how construction field crews perform tasks on construction jobsites. It includes three context types: “task”, “schedule”, and “equipment and tools”.

6) Task context is a general description of what field crews need to accomplish.
7) Schedule context specifies at what time a task was/is/will be executed by field crews.
8) To accomplish certain tasks, construction field crews need to know what equipment and tools to use, and if the equipment and tools are available and in good condition.

3.4 External factors context

External factors context provides information beyond the crew level. It includes “weather”, “work zone”, “site condition”, “nearby object”, “technical and safety specification”, “cultural and legal constraint”, “industry policy”, “public relation”, and “computing entity”.

9) Weather context considers temperature, precipitation, humidity, etc.
10) Work zone context provides information about the space where construction field crews perform tasks on site. More specifically, it includes information about absolute locations, relative locations, and interdependent relationships among different components.
11) Site condition context refers to the condition of the land where a construction site is located. It includes the physical, geophysical, environmental, ecological, geotechnical, and archaeological conditions.

12) Nearby object context refers to the presence of stacking materials, equipment, tools, components, openings, etc., which are useful for a task but could prevent crew members from performing their tasks as planned.
13) Technical and safety specification context specifies the requirements that construction field crew members should meet to avoid technical and safety incidents while performing tasks.
14) Cultural and legal constraint context considers information including language barriers, cultural norms, and contractual requirements that constrain field tasks.
15) Industry policy context considers requirements from regulatory agencies relevant for the construction field work.
16) Public relation context refers to information that is associated with the neighboring community.
17) Computing entity context specifies information about computing devices, the internet, etc.

4. The Construction Context Information (CCI) model

We built a CCI model based upon the validated 17 basic context types using the ontology-based modeling method. The model defines attributes of these context types and their relationships. Encoded in OWL, the CCI model provides a formal and machine-interpretable representation of field context. The main ontology structure lays out all context types. The following are sub-models for the four categories of the field context (i.e., product, organization, process, and external factors). Sub-models indicate that the CCI model can be extensible for a specific sub-domain of interest. To illustrate our modeling concepts, this section starts with a typical jobsite task scenario, which shows how construction field crews interact with field context.

4.1 A task scenario on construction jobsites

John is a crew member who does concrete work. Every morning, he gets instructions of what task to do for the day from his superintendent. On Monday morning, John is assigned to install a precast concrete wall (i.e., the task context) with three other crew members (i.e.,

the crew composition and crew member context) on the east side of the jobsite on the ground level (i.e., the work zone context). The task is scheduled to start at 7AM and finish at 8AM (i.e., the schedule context). When preparing for the task in the morning, John checks the status of the grouting material, precast concrete wall, crane, and supporting frames (i.e., the material, component, equipment and tools context). The delivery of the precast wall is delayed in the morning for 60 minutes. Moreover, John finds that the work zone is not accessible because unused equipment is occupying the space. After moving the unused equipment out of the way, the crew starts mobilization; however, by now, the start time is delayed by 30 minutes. During the installation, it is found that the supporting frame is not of the right size. Hence, they need to stop and change to the right frame (i.e., change of the equipment and tools context), which takes an extra 45 minutes. In the afternoon, it starts to rain (i.e., the weather context). Water is all over the site and the ground becomes muddy (i.e., the site condition context). All tasks are halted in the afternoon. For every task, the concrete crew must, of course, strictly follow relevant industry policy and technical and safety specifications. One crew member William is found not wearing protection glasses, so he is warned and asked to wear protection glasses before he can return to work.

Through the description of this task scenario, we show that a context type is characterized by attributes (e.g., availability of the component context). In addition, we illustrate relationships between context types; for example, a task requires materials, components, equipment and tools. Attributes of and relationships between context types were extracted from field task scenarios and then specified with the CCI model.

4.2 The top-level context ontology structure

Figure 5 shows the top-level ontology structure of the CCI model. As the figure shows, the field context includes four sub-domains, each of which consists of corresponding basic context types as discussed in the previous section. The OWL built-in property rdfs:subClassOf was used to define “subclass” relations between the four context sub-domains and specific context types. It allows for hierarchically structuring sub-class context types.

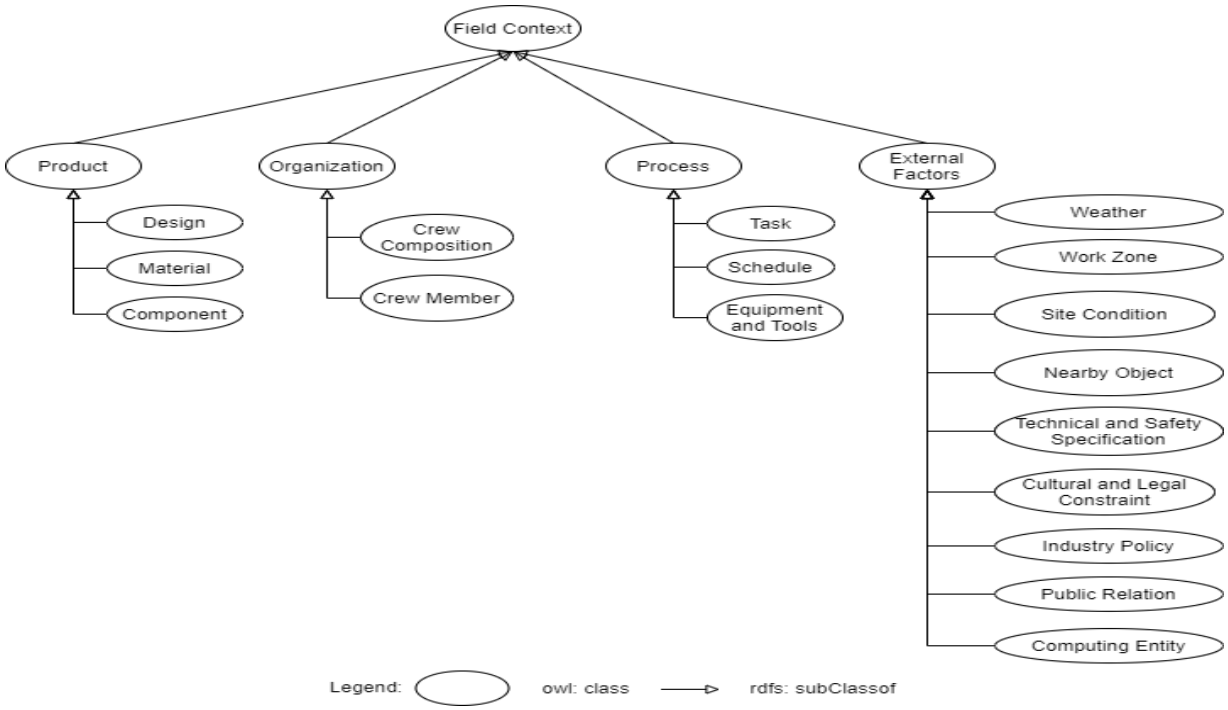

Figure 5. Top-level context ontology structure

Given the top-level ontology structure and the 17 context types, their relationships with each other (represented in owl: ObjectProperty) were defined, as shown in Figure 6, based on our knowledge of construction field work. We defined these relationships based on our analysis of 152 instances of field task scenarios that we observed from five projects, including a mixed-use commercial and residential building, an office building, a medical facility, and a concrete foundation. Undoubtedly, there are other relationships, which can be added easily as necessary. We wanted to define these relationships to enable reasoning about context. Specifically, for instance, a relation "isPerformedBy" was defined to show that a task should be performed by a crew member, and a relation "hasDesign" was defined to show that a task should be performed according to the design so that the crew can deliver the product as requested. More examples of the relationships include: a "Task" has relations, including "isAffectedBy", "use", "isConstrainedBy", "hasWeather", "isGuidedBy", "hasSchedule", "hasEquipmentAndTool", "hasCrewComposition", "is-PerformedBy", "hasMaterial", "hasDesign", "hasComponent", "isLocatedIn", "hasSiteCondition", "isFollowerOf" and "isPrecedentOf", with other context types. It

should be noted that the upper ontology structure generally specifies the relationships between the 17 context types, while for each sub-domain (i.e., product, organization, process and external factors), a sub-ontology (i.e., product ontology, organization ontology, process ontology and external factors ontology) was built to show the details of the corresponding context types and their attributes (represented in owl: DatatypeProperty). For each context type, we identified at least one attribute so that we were able to perform reasoning.

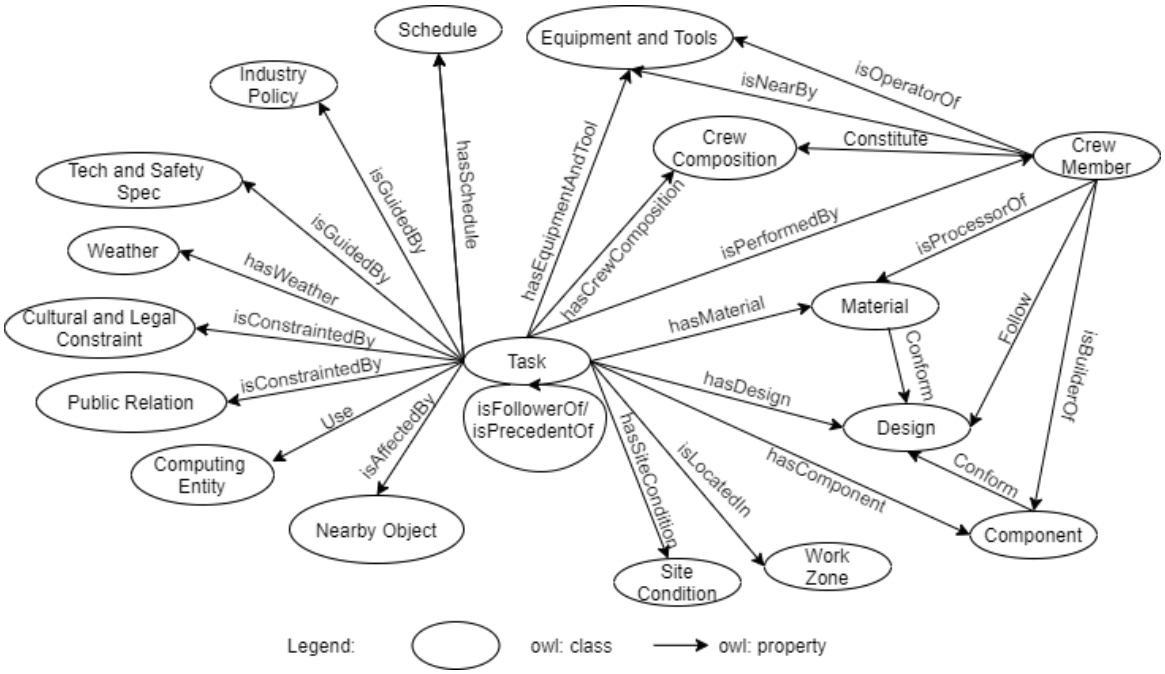

Figure 6. Relationships of context types

4.3 Product ontology

The product ontology shows the attributes associated with “design”, “component”, and “material” (Figure 7). For example, a design has “dimensions” (e.g., “height”, “width”, “length”), “constraints” (e.g., “dependency” and “connection”), “quantity”, different “versions”, etc. An example of data values is that the height of a precast wall is six feet. The attribute “version” indicates that a design can be updated and when there are changes to the design, a new version of the design will be created. Materials and components are assigned a “quantity” and “dimensions”. By comparing the information of components and materials with the design information, it could be inferred whether there are conflicts of

product information. Moreover, a component usually has an “identification number” to help field crews check if it is the right one to be installed. As to the component and material context, they both have a relation “hasSchedule” to connect with the schedule context to indicate the temporal information (e.g., the delivery time).

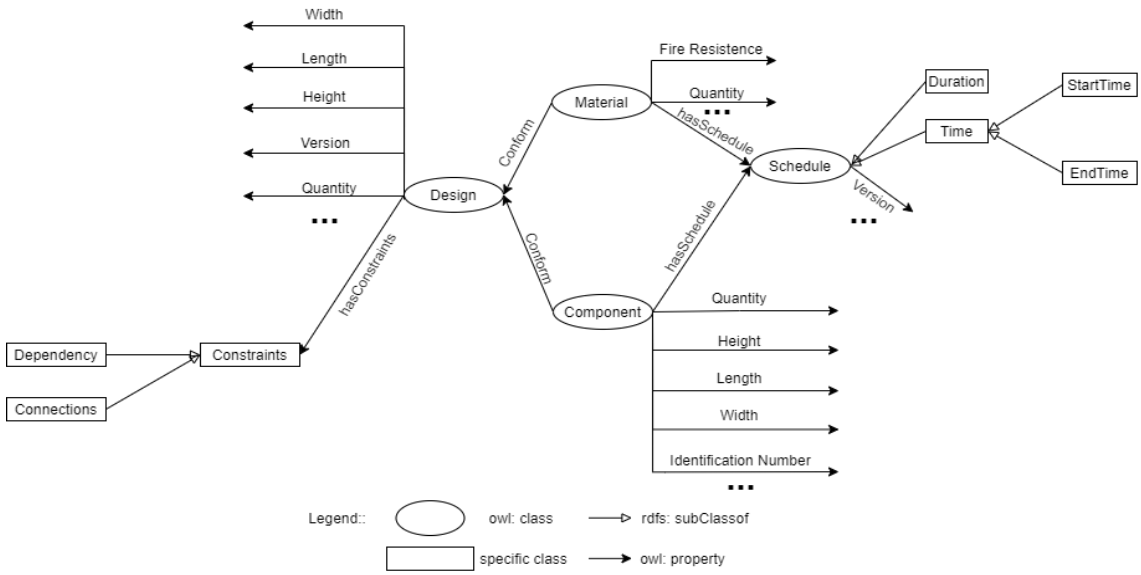

Figure 7. Product ontology

4.4 Organization ontology

The organization ontology, as shown in Figure 8, defines attributes associated with “crew composition” and “crew member”. Crew composition includes “size” to indicate the number of crew members. For example, the size of a concrete crew has a data value of 6. A crew member has his/her “profile” (e.g., “name”, “job title”, “age”, “gender”, “email”, and “phone number”) and “action” (e.g., “idling”, “working”) information.

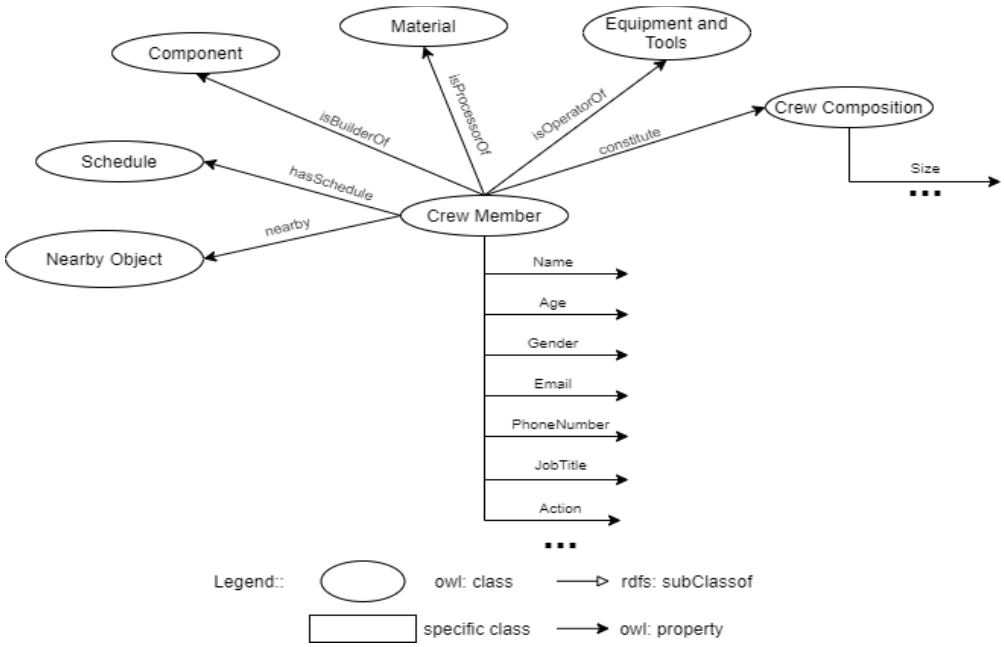

Figure 8. Organization ontology

4.5 Process ontology

The process ontology, as shown in Figure 9, defines attributes of “task”, “schedule”, and “equipment and tools”. A task should have a description to clearly describe what is required to be accomplished by the field crew. Moreover, a task is associated with different stages, including “gettingReady”, “inProgress”, “inspection”, “rework”, and “complete”. In addition, a task has its status (e.g., “delay” and “onSchedule”) to indicate whether it is proceeding as planned.

For “schedule”, it has “time” (e.g., “startTime” and “endTime”) and “version”. For example, the “startTime” of install a precast wall has a data value of 8AM. Since there may be changes to the “schedule”, “version” is the attribute to indicate when a new “schedule” is created. As to “equipment and tools”, it has attributes including “availability”, “access”, “condition”, and “electricity”.

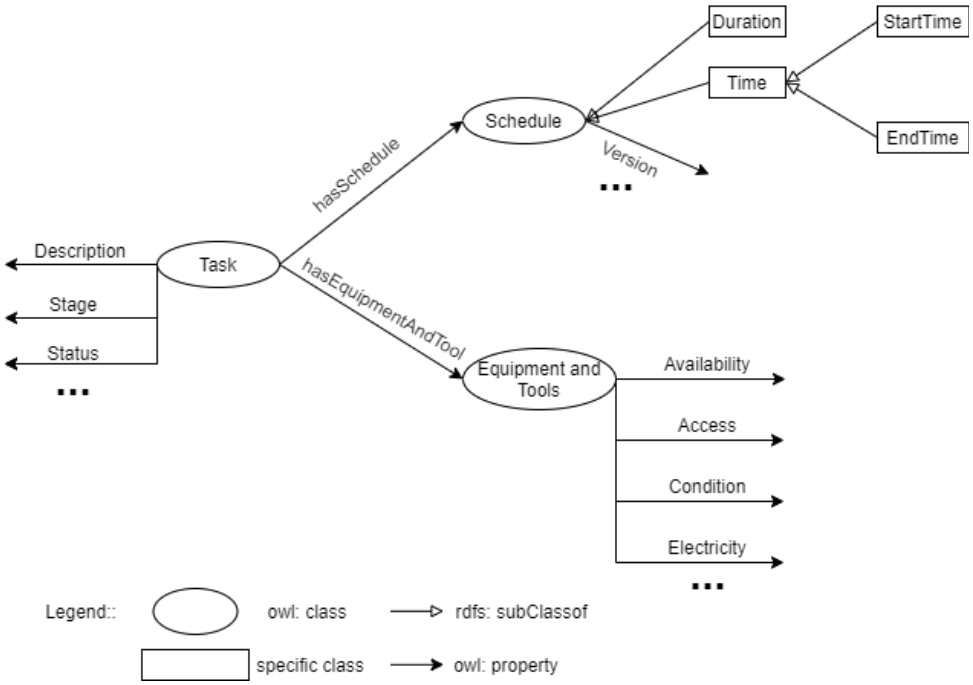

Figure 9. Process ontology

4.6 External factors ontology

The external factor ontology, as shown in Figure 10, shows attributes of “weather”, “work zone”, “site condition”, “nearby object”, “technical and safety specification”, “cultural and legal constraint”, “public relation”, “industry policy” and “computing entity”.

The “weather” context has attributes including “humidity”, “temperature”, “wind”, and “precipitation”. “Work zone” context has “access” and “space”. “Site condition” context consists of “contaminated soil”, “power line”, and “underground water”. “Nearby Object” context includes “hazard” and “occupied space” to indicate the presence of things that can be hazardous to field crew members or occupy spaces. As to “computing entity”, its implementation requires “training” and “connectivity”. For “technical and safety specification”, “cultural and legal constraint”, “public relation”, “industry policy” and “computing entity”, attributes include “description” and “version”.

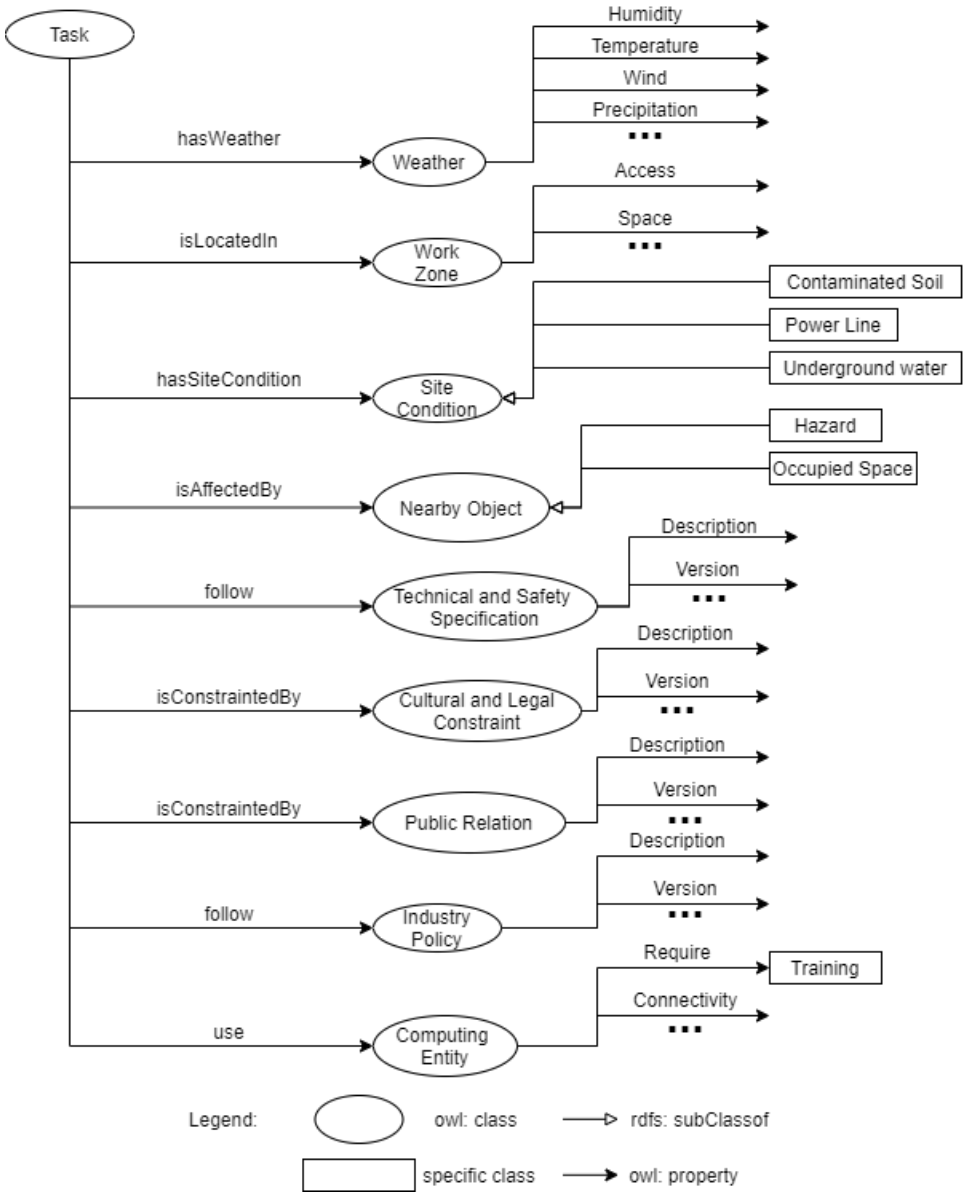

Figure 10. External factors ontology

4.7 Summary

To summarize, we defined relationships between the validated 17 context types and their attributes based on analysis of context information associated with 152 field task scenarios from five projects. There may be other relationships and attributes, but they could be easily defined and added with the same approach.

The CCI model was represented using OWL concepts. Each context type was represented as an owl:Class. Attributes of context types were represented using owl:DatatypeProperty (e.g., availability of the material context). As to the value of an attribute, for example, the availability of the material context has a value of "No". Relationships between context types were represented using owl:ObjectProperty (e.g., the "hasMaterial" relation between the material context and the task context). The OWL-encoded ontology modeling method makes the model machine-interpretable, and therefore it could be used to develop context-aware systems that help observe, interpret, reason about, and communicate field context information.

5. Context reasoning

This section shows how reasoning about context information was performed based on the CCI model. The CCI model was built using OWL. However, OWL has reasoning limitations (O'Connor et al., 2005). For example, it cannot deal with situations where context is missing or ambiguous and it does not support general rules (e.g., the "If . . . Then . . . Else" rule). Therefore, an OWL-based context model can reason about implicit but meaningful context information if combined with general rules. Implementation of user-defined general rules makes it more flexible for an OWL-based context model to perform reasoning. To add general rules to the OWL-based context model, the Semantic Web Rule Language (SWRL) can be implemented (Bae et al., 2013). A SWRL rule is composed of an antecedent and a consequent, both of which is composed of several atoms. The antecedent specifies conditions that needs to be evaluated. A rule specifies that if the atoms in the antecedent are true, the atoms in the consequent should also be true (Horrocks

et al., 2004). Rules were written with OWL concepts (e.g., classes, properties, individuals, and literals).

SWRL rules add expressiveness and completeness to an OWL-based context model because user-defined rules enhance the expressive capability and allow inferences over OWL individuals, so implicit but meaningful context information associated with these individuals can be obtained. By incorporating rule-based constraints, the OWL-based ontology derives high-level context and detects inconsistencies of context; therefore, it can be more applicable to the dynamic construction environment.

To further illustrate how context reasoning can be performed based on the CCI model, we use real construction scenarios to show how high-level context, such as "the status of a task", can be deduced from low-level context through the creation of reasoning rules**.** For example, in the "gettingReady" stage, the status of a task will be derived to be "delay" if a crew member is working too close to an opening or if there are not enough crew members; in addition, conflicts between the design and component delivered to site will delay a task. In the "inProgress" stage, unfavorable weather can halt a task and change the task status to "delay"; for example, if it rains, field crews have to stop their task. In the "inspection" stage, the task cannot be considered as complete if the finished component does not meet design specification and therefore the status of the task is inferred to be "delay". Table 7 shows examples of general reasoning rules that we formulated to show how context reasoning were performed.

Table 6. Examples of reasoning rules

Stage	Status	Reasoning Rules
gettingReady	Delay – due to lack of crew members	Crew Composition(?x) ^ hasTask(?x, installPrecastWall) ^ (?x size ?s) ^ swrlb:lessThan(?s, 3) ^ (installPrecastWall stage gettingReady) => (installPrecastWall status Delay)
gettingReady	Delay – due to potential safety accident	Crew Member(?x) ^ hasTask(?x, installEquipment) ^ Work Zone(?y) ^ locatedIn(installEquipment, ?y) ^ nearby(?x, opening) ^ (installEquipment stage gettingReady) => (installEquipment status Delay)
gettingReady	Delay – due to conflicting information	hasDesign(installPrecastWall, design) ^ (design height h_1) ^ hasComponent(installPrecastWall, PrecastWall) ^ (PrecastWall height h_2) ^ swrlb:notEqual(?h_1, ?h_2) ^ (installPrecastWall stage gettingReady) => (installPrecastWall status Delay)
inProgress	Delay – due to unfavorable weather	Crew Member(?x) ^ hasTask(?x, installPrecastWall) ^ affectedBy(installPrecastWall, Rain) ^ (installPrecastWall stage inProgress) => (installPrecastWall status Delay)
inspection	Delay – due to finished work that does not meet design specification	hasDesign(pourConcreteFooting, design) ^ (design thickness t_1) ^ hasComponent(pourConcreteFooting, ConcreteFooting) ^ (ConcreteFooting thickness t_2) ^ swrlb:notEqual(t_1, t_2) ^ (pourConcreteFooting stage inspection) => (pourConcreteFooting status Delay)

6. Conclusions

To provide construction field crews with context information and its changes, a formal representation of field context is necessary. This paper shows that it is not only essential to discern different types formally, but also to interpret and reason about field context information. Therefore, we had to understand attributes of and relationships between different context types to enable interpretation and reasoning. Our observation and analysis of 152 field task scenarios from five construction projects equipped us with such an understanding.

In this paper, we presented a Construction Context Information (CCI) model built upon 17 validated context types, to formally represent field context. The CCI model defines relationships between these context types and their associated attributes. The CCI model is in the form of an ontology encoded in OWL, a machine-interpretable language; therefore, the CCI model supports the development of context-aware systems, which enhance field communication. Coupled with reasoning rules, the CCI model was shown to help detect

conflicts of context information and infer high-level context information (e.g., the status of a task) based on low-level context information (e.g., weather).

For future work, we will explore how on-site communication could be improved by implementing mobile computing technologies based on the CCI model. A lack of context-awareness hampers field crew members from getting informed of context information and its changes in a timely manner. As a result, field tasks could be delayed and other issues might arise (e.g., safety incidents). To approach this problem, our attention will be on whether context-based instructions could be informative for field crews and how mobile computing technologies could be leveraged to provide context-based instructions on construction jobsites. Moreover, considering that the CCI model could be the basis for different context-aware systems on construction jobsites, we suggest that the interoperability among different context-aware systems should be studied.

Acknowledgement

This study was funded by the Seed Funding provided by the Center for Integrated Facility Engineering (CIFE) at Stanford University.

References

Acar, E., Kocak, I., Sey, Y. and Arditi, D., 2005. Use of information and communication technologies by small and medium-sized enterprises (SMEs) in building construction. Construction Management and Economics, *23*(7), pp.713-722.

Adriaanse, A., Voordijk, H. and Dewulf, G., 2010. Adoption and use of interorganizational ICT in a construction project. *Journal of construction engineering and management*, *136*(9), pp.1003-1014.

Anumba, C.J., Pan, J., Issa, R.R.A. and Mutis, I., 2008. Collaborative project information management in a semantic web environment. Engineering, Construction and Architectural Management.

Arruda, L., França, S. and Quelhas, O., 2014. Mobile Computing: Opportunities for Improving Civil Constructions Productivity. *International Review of Management and Business Research*, *3*(2), p.648.

Aziz, Z., Anumba, C.J., Ruikar, D., Carrillo, P. and Bouchlaghem, D., 2006. Intelligent wireless web services for construction—A review of the enabling technologies. *Automation in Construction, 15*(2), pp.113-123.

Bae, H., Golparvar-Fard, M. and White, J., 2013. High-precision vision-based mobile augmented reality system for context-aware architectural, engineering, construction and facility management (AEC/FM) applications. *Visualization in Engineering*, *1*(1), pp.1-13.

Bettini, C., Brdiczka, O., Henricksen, K., Indulska, J., Nicklas, D., Ranganathan, A. and Riboni, D., 2010. A survey of context modelling and reasoning techniques. *Pervasive and mobile computing*, *6*(2), pp.161-180.

Bikakis, A., Patkos, T., Antoniou, G. and Plexousakis, D., 2007, November. A survey of semantics-based approaches for context reasoning in ambient intelligence. In *European Conference on Ambient Intelligence* (pp. 14-23). Springer, Berlin, Heidelberg.

Bowden, S., Dorr, A., Thorpe, T. and Anumba, C., 2006. Mobile ICT support for construction process improvement. *Automation in construction*, *15*(5), pp.664-676.

Brickley, D., Guha, R.V. and McBride, B., 2014. RDF Schema 1.1. *W3C recommendation*, *25*, pp.2004-2014.

Chen, G. and Kotz, D., 2000. A survey of context-aware mobile computing research.

Fathi, M.S., Abedi, M., Rambat, S., Rawai, S. and Zakiyudin, M.Z., 2012. Context-aware cloud computing for construction collaboration. *Journal of Cloud Computing*, *2012*, p.1.

Horrocks, I., Patel-Schneider, P.F., Boley, H., Tabet, S., Grosof, B. and Dean, M., 2004. SWRL: A semantic web rule language combining OWL and RuleML. *W3C Member submission, 21*(79), pp.1-31.

Kasim, N.B. and Ern, P.A.S., 2010. The awareness of ICT implementation for materials management in construction projects. *Int. J. of Computer and Communication Technology*, *2*(1), pp.1-10.

Kasim, N., 2011. ICT implementation for materials management in construction projects: case studies. *Journal of Construction Engineering and Project Management*, *1*(1), pp.31-36.

Lam, P.T., Wong, F.W. and Tse, K.T., 2010. Effectiveness of ICT for construction information exchange among multidisciplinary project teams. *Journal of Computing in Civil Engineering*, *24*(4), pp.365-376.

McGuinness, D.L. and Van Harmelen, F., 2004. OWL web ontology language overview. *W3C recommendation, 10*(10), p.2004.

Nurmi, P. and Floréen, P., 2004. Reasoning in context-aware systems. *Helsinki Institute for Information Technology, Position paper*.

O'connor, M., Knublauch, H., Tu, S. and Musen, M., 2005. Writing rules for the semantic web using SWRL and Jess. *Protégé With Rules WS, Madrid*.

Omar, B. and Ballal, T., 2009. Intelligent wireless web services: context-aware computing in construction-logistics supply chain. *ITcon, 14*(Specia), pp.289-308.

Perkinson, C.L., Bayraktar, M.E. and Ahmad, I., 2010. The use of computing technology in highway construction as a total jobsite management tool. *Automation in Construction*, *19*(7), pp.884-897.

Perera, C., Zaslavsky, A., Christen, P. and Georgakopoulos, D., 2013. Context aware computing for the internet of things: A survey. *IEEE communications surveys & tutorials*, *16*(1), pp.414-454.

Reichle, R., Wagner, M., Khan, M.U., Geihs, K., Lorenzo, J., Valla, M., Fra, C., Paspallis, N. and Papadopoulos, G.A., 2008, June. A comprehensive context modeling framework for pervasive computing systems. In IFIP International Conference on Distributed Applications and Interoperable Systems (pp. 281-295). Springer, Berlin, Heidelberg.

Sattineni, A. and Schmidt, T., 2015. Implementation of mobile devices on jobsites in the construction industry. *Procedia Engineering*, *123*, pp.488-495.

Strang, T. and Linnhoff-Popien, C., 2004, September. A context modeling survey. In *Workshop Proceedings*.

Studer, R., Benjamins, V.R. and Fensel, D., 1998. Knowledge engineering: Principles and methods. *Data & knowledge engineering, 25*(1-2), pp.161-197.

Wang, S., Wan, J., Li, D. and Liu, C., 2018. Knowledge reasoning with semantic data for real-time data processing in smart factory. *Sensors*, *18*(2), p.471.

Ye, J., Coyle, L., Dobson, S. and Nixon, P., 2007. Ontology-based models in pervasive computing systems. *The Knowledge Engineering Review, 22*(4), pp.315-347.

Yuan, B. and Herbert, J., 2014. Context-aware hybrid reasoning framework for pervasive healthcare. *Personal and ubiquitous computing, 18*(4), pp.865-881.

Chapter 4: Context-based Work Instructions for Construction Field Tasks

Abstract: Effective communication is critical for construction field crews in terms of helping control task cycle time. However, today's methods don't offer systematic awareness of field context to construction field crews. State-of-the-art work instructions do not transmit context information in a just-in-time way nor respond to changes of context. To resolve this issue, this research proposes to enhance context-awareness for field crews through informing field crews of context information and its changes. Thanks to the emergence of mobile computing technologies, coupled with more affordable devices, it becomes possible to provide and update field context information in a just-in-time manner. This paper presents a new form of work instructions based on field context for field crews to guide the execution of construction field tasks. The proposed work instructions have two distinguishing features: informed of changes and just-in-time. A mobile-based prototype was developed to prepare, distribute, and update the proposed context-based instructions for construction field crews. We conducted a field study on an office renovation project to evaluate whether and how context-based work instructions affect the cycle time of field tasks. This controlled experiment was conducted on 14 pairs of treatment and control groups in that context-based work instructions were provided to the treatment group while not to the control group. The results show that, for the same task, the actual cycle time in the treatment group varies less than in the control group. In addition, the treatment group generally has a much smaller delta between the actual and the planned cycle time than the control group. The averaged percent deviation of the treatment group, calculated by dividing the delta by the planned cycle time, is 35.7% smaller than that of the control group for these 14 pairs. Overall, the proposed context-based work instructions have been shown to reduce the variability of the cycle time of field tasks in two ways: less variation in the actual cycle time of the same types of tasks and improved alignment between the actual and planned cycle time.

Key words: Work Instruction, Field Context, Change Management, Just-in-Time, Last Planner System, Cycle Time

1. Introduction

Effective communication is critical for construction field work (Koo and Fischer 2000, Haymaker et al., 2000). From the value stream perspective, construction field crews are those who directly add value onto the final building product. Therefore, how they perform field tasks affects the project performance to a considerable extent. Although many efforts (e.g., information management systems, expert systems, etc.) have been made towards boosting communication among different stakeholders of construction projects, research focused on the information passed onto or reported by construction field crews has been limited. Mourgues and Fischer (2008) put forward a method to generate field instructions that provide necessary information for field crews. Their instruction template is considered the main point of departure for this research. However, their method was confined by the available technologies at the time. As a result, the instructions could be outdated or irrelevant due to the lack of dealing with changes of field context in a timely manner.

To improve communication with construction field crews, we first analyzed criteria on field work instructions and identified two primary criteria: informed of changes and just-in-time. In line with these two criteria, we suggested that work instructions should adapt to changes in field context so that the instructions provide information about changes in field situations in a timely manner. As mobile devices have become more affordable and found their way to construction jobsites, mobile computing has the potential to be utilized to enhance context-awareness for construction field crews. As such, we developed a mobile-based prototype by which context-based work instructions can be prepared, distributed, and updated. The proposed context-based field instructions were evaluated through a case study on a construction project to analyze whether field crews working with work instructions that update them about changes in context perform more predictably than crews that use work instructions without context updates. In particular, we wanted to find out whether task cycle time would vary more or less and whether actual task cycle times would be closer to planned cycle times or not.

The organization of this paper is as follows: section 2 discusses the motivation for this research; section 3 presents an analysis of how lacking awareness of field context

information and its changes affected the variability of task cycle time through a preliminary case study; section 4 presents related work to this research, including criteria on work instructions, change management for construction field tasks, just-in-time principles in construction and the Last Planner System; section 5 presents the research method, which discusses the collection of the field context, development of a mobile-based prototype to facilitate communication of work instructions and implementation of the context-based work instructions in a case study; section 6 discusses the cast study results, and section 7 draws conclusions of this paper.

2. Motivation

This research started with an analysis of existing work instructions. We analyzed two types of work instructions: (1) oral work instructions, which are common practice for construction work; and (2) existing advanced work instructions, which are more formal and are applied by sophisticated construction companies.

Although oral instructions are common on construction jobsites, they are usually error-prone and often do not provide sufficient information to guide field crews. In comparison, existing advanced work instructions have some advantages, such as inclusion of illustrative and informative diagrams. Figure 11 shows an example of advanced work instructions. This work instruction is illustrative by including 3D and 2D views of building objects. However, it still lacks necessary information. For example, based on what is presented in the instruction, it does not include procedures which field crews should follow to install the rebar. It might not be a problem for experienced field crew members, but what if the procedures have to be changed due to changes in the work zone? In other words, this advanced instruction does not inform the changes neither. To give more examples of possible changes to field context, what if unexpected site conditions, which hinder the installation, appear? Or what if the rebar delivered does not conform with the specifications? Or what if the installation requires special safety protection which the field crews are unaware of? These are possible scenarios, for which the displayed instruction does not contain necessary context information nor reflect its changes in a just-in-time manner.

Field crews might have to halt the task when running into such scenarios, which could lead to differences between the expected or planned and actual task cycle times.

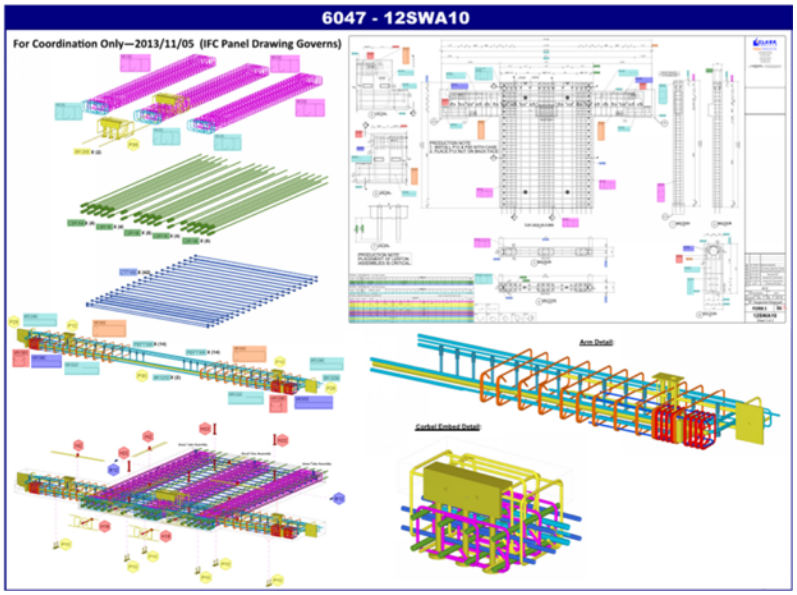

Figure 11. An example of existing advanced work instruction (Sourced from Clark Pacific)

3. Preliminary case study

Before designing a detailed study on the impact of context information as part of work instructions on task cycle time, we carried out a preliminary study to determine whether work instructions are likely to impact task cycle time. To investigate how the task cycle time might vary when field crews are not sufficiently informed of context information and its changes in a just-in-time way, we conducted a first-run study on a precast parking structure project in three steps:

1) We recorded time-lapse videos of the installation of precast components (e.g., shear walls, columns). The videos were later analyzed to calculate the cycle time of each installation.
2) Meanwhile, we took notes of problems that were associated with each installation. For example, after mobilizing to install a precast shear wall, the field crew found that the rebar of the precast shear wall was too long to fit into the hole on the ground.

Therefore, they spent extra time cutting the rebar and the actual cycle time for the installation got extended.

3) Finally, we made a correlation analysis between the cycle time of each installation and the observed problems. Moreover, for the same type of precast component, the cycle time of installations with problems observed versus those without any problems was compared.

Installations of precast shear walls on the ground level, for instance, were observed to be completed with varied cycle times. In Figure 12, the number of each shear wall in the figure represents the installation sequence. The cycle time started from 111 minutes for the first installation, however it decreased to 23.5 minutes to the fourth. However, the cycle time increased again. Overall, the cycle times of the installations, for the same size of shear wall installed by the same crew, varied considerably.

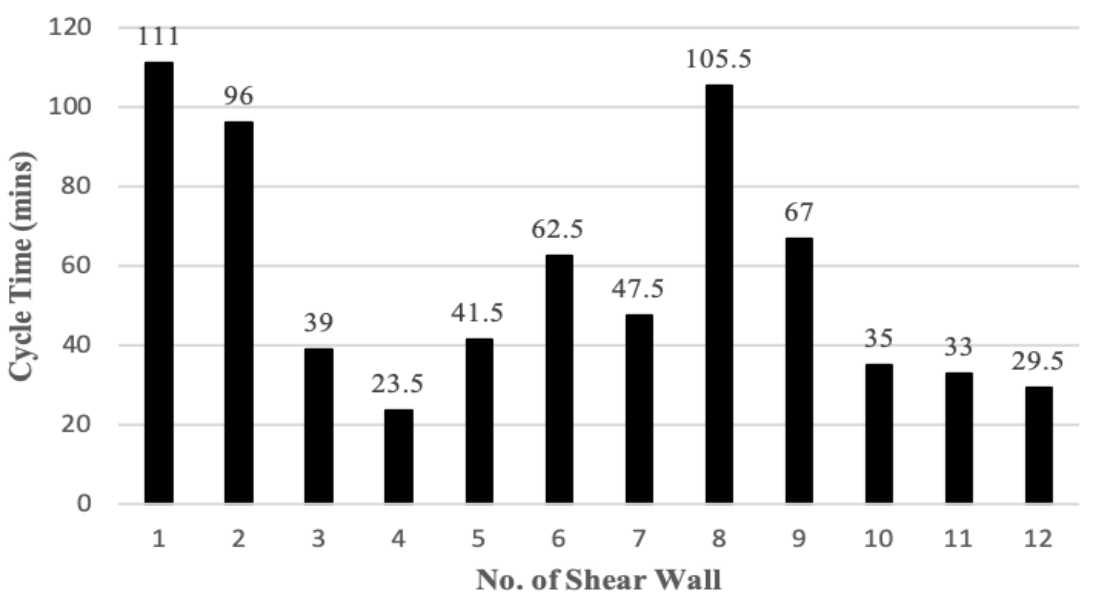

Figure 12. Cycle times of observed shear wall installations on a precast project

Figure 13 shows the average cycle time for installing shear walls with problems observed (7) versus those without problems observed (5). The problems led to a significant increase in the cycle time of installation. Further analysis of the problems showed that the problems could be attributed to the lack of awareness of the field context and its changes in a timely manner. For instance, Figure 14 is a photo showing that field crews attempted to install a precast shear wall. A mismatch of the rebars beneath the shear wall with the cast-in-place cups (holes) increased the cycle time for the installation. Field crews assumed that they got the precast shear wall exactly as designed, however, extra adjustments were needed, which led to delays in the installation cycle.

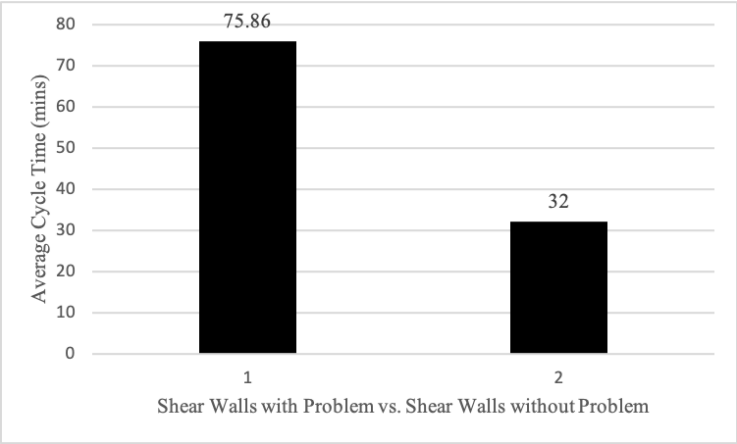

Figure 13. Production cycle time of shear walls on the ground level

Figure 14. Site photo: installation of precast shear wall

Thereafter, we performed the same analysis over other shear walls and precast columns, beams, and spandrels. It was found that the installation of about 40% of the shear walls,

33% of the columns, and 25% of the spandrels were negatively affected by the lack of awareness of field context and its changes in a timely manner.

Through this first-run analysis, we found that when construction field crews were not aware of field context information and its changes, it was likely that the task cycle time could vary significantly. Moreover, the analysis implies that, if field crews could check and verify the context ahead of time, they might have been able to report changes of context, if any, to foremen/superintendents and react accordingly. Therefore, we were curious about whether work instructions that include field context and communicate its changes could be helpful for field crews in terms of achieving the task planned cycle time.

4. Related work

This research aims to improve communication on construction jobsites via providing field crews with field work instructions. To understand what makes high-quality work instructions, we reviewed the criteria from both inside and outside the construction domain. The exploration led us to create work instructions that are informed of changes and communicated in a just-in-time manner. Hence, we reviewed research regarding change management on construction jobsites and specifically about how to mitigate negative impacts of changes on construction field work. Moreover, the implementation of the proposed work instructions should be in line with short-cycle planning (e.g., weekly planning) so as to satisfy the "just-in-time" feature and provide actionable information for field crews. Therefore, prior studies regarding just-in-time principles in construction and the Last Planner System were also reviewed. Lastly, we reviewed context-aware applications in the domain of construction.

4.1 Criteria for work instructions

In order to create work instructions for construction field crews, it is important to first identify the criteria. Some researchers have conducted comprehensive reviews on characteristics of work instructions for construction workers. Gorse and Emmitt (2003) pointed out that work instructions should be clear, concise, complete, error-free,

meaningful, relevant, accurate, and timely to the workers. Mourgues and Fischer (2008) summarized previous studies and put forward a list of characteristics of work instructions. According to their study, work instructions should be written, complete, self-contained, specific, formal, dynamic, explicit, concise, and illustrative. 'Self-contained' means that construction workers would not need to refer to any other documents or drawings when given work instructions, and 'concise' means that work instructions would only show critical details but leave out general knowledge. These features reflect the basic requirements for formulating high-quality work instructions. More specifically, the criteria can be categorized into three aspects of work instructions: content, format, and delivery.

From the content perspective, LeFevre and Dixon (1986) stated that instructions should avoid being too generic. For field crews, work instructions need to be specific to field tasks. Di Eugenio (1992) pointed out that goal setting could impact the interpretation of the instructions and could, therefore, affect the execution of field tasks. Therefore, goals should be explicitly stated in field work instructions. Mourgues and Fischer (2008) developed a field instruction template which met these criteria to provide task-specific information and set explicit goals. However, their instruction could not inform field crews of changes to the information.

In terms of the format of work instructions, researchers have shown that illustrative diagrams are necessary to be included. Austin et al. (1995) stressed that illustrations could help workers develop a better understanding of work instructions because information could be more clearly conveyed with diagrams as references. Heiser et al. (2004) stated that diagrams are integral to instructions for spatial tasks such as assembly. Furthermore, Watson et al. (2010) explored the effectiveness of different formats of instructions, and they verified that dynamic instructions could result in shorter build time than static instructions with equivalent amount of information. Dynamic instructions are more explicit because they can show the sequence of tasks. In addition, according to Heiser et al. (2004), work instructions should be expressed clearly in a step-by-step way, and diagrams should only be used for major steps. Based on prior research, diagrams, or dynamic diagrams in particular, should be included in work instructions for illustrative purposes. In addition,

different types of diagrams have different functions. Heiser et al. (2004) categorized diagrams into three types: structural diagrams, action diagrams, and part menus. Structural diagrams show two or more parts in configured positions. Action diagrams demonstrate the necessary actions to join one part to another. Part menus depict the characteristics of a part. Different diagrams vary in terms of demonstrating the instructions. In this case, a combination of the three types of diagrams is recommended and would be helpful in terms of effectively conveying the information. Li et al. (2004) proposed a 2.5D layer-based diagram representation method, which provides users with authoring tools for quickly creating layered diagrams. The diagram representation is interactive in that it enables users to dynamically filter the information as they need. Another illustration diagram is called cutaway. Li et al. (2007) put forward a system for users to author interactive cutaway illustrations of complex 3D models. The system enables users to directly cut as they like and show the internal parts of an object. Even though the above-discussed research was done for assembly in manufacturing, the idea of applying diagrams to convey information to workers can be applied to the construction industry. In fact, considering that the construction environment is more volatile than manufacturing, inclusion of diagrams and detailed task breakdowns could make work instructions more explicit. For example, Gao and Fischer (2006) suggest color drawings to improve communication with instructions. In the instruction template provided by Mourgues and Fischer (2008), a 2D floor plan is included to illustrate the design. In addition, their instruction template includes a description of key procedures to show how a task should be performed. However, the instruction was not designed to be interactive for field crews and, as such, crews do not have a way to report changes they run into at the workface.

As to the timing of work instructions, Antifakos et al. (2002) suggest just-in-time instructions for furniture assembly which adapt to states of the assembly task so that the instructions can be relevant to what the workers are doing. Just-in-time instructions can be applicable to the construction industry so as to timely inform field crews of field context information and its changes. At the same time, however, it is the varied construction environment that makes just-in-time instructions hard to create. Mourgues and Fischer (2008) argued that, for the time being, work instructions should be delivered complete to

construction workers before they perform tasks due to the difficulty in creating just-in-time instructions especially in the construction environment. In this paper, we argue that, with the development of mobile computing technologies and more affordable devices, just-in-time instructions become feasible for construction field tasks.

While previous research has provided insights into high-quality work instructions in both the manufacturing and construction industry, this paper emphasizes two extended criteria of high-quality work instructions for construction field crews: informed of changes and just-in-time communication. The first feature, "informed of changes", means that work instructions should inform field crews of changed context and get updated to respond to the changes. Changes of context are either passed down from context sources or reported from field crews at the workface. The second feature, "just-in-time communication", means that communication of field context information and its changes is in a timely manner.

4.2 Management of changes that affect field crews' work

Management of changes for construction field work is an important aspect of project management (Motawa et al., 2007). Changes are unchangeable factors that could lead to failure of tasks and negatively affect the schedule (Nasir et al. 2003). Construction projects often involve multiple feedback processes to deal with changes and their impact on the construction performance (Sterman, 1992; Othman et al., 2004). However, previous research on work instructions barely considers changes, neglecting that work instructions can easily be irrelevant due to the dynamic construction environment. This study took Motawa et al.'s view on the life cycle of changes (Motawa et al., 2007). They identified two types of feedback processes: balancing and reinforcing. The balancing process resolves changes with proper control actions implemented while the reinforcing process may generate other changes when incorrect control actions are taken. Those two types of processes are triggered depending on the availability of relevant information for construction personnel. Regarding the balancing process, for instance, when a field crew is assigned to install a precast column, if they are aware ahead of time that part of the installation spot is occupied by a cast-in-place concrete component, the crew can find a

solution to the situation and does not affect succeeding tasks. As to the same situation, however in the reinforcing process, if field crews do not realize it until they start working, the installation cycle will be delayed and the schedule of succeeding tasks is negatively affected. We propose to use context-based work instructions to provide timely updates about changes of context so that field crews can integrate these changes to make reactive actions. In other words, field crews can have more balancing processes but avoid reinforcing processes. Figure 15 illustrates an update of Motawa et al.'s feedback processes with context-based work instructions being implemented. Incorporated with changes and corresponding reactive actions, context-based work instructions could help field crews adapt to construction dynamics.

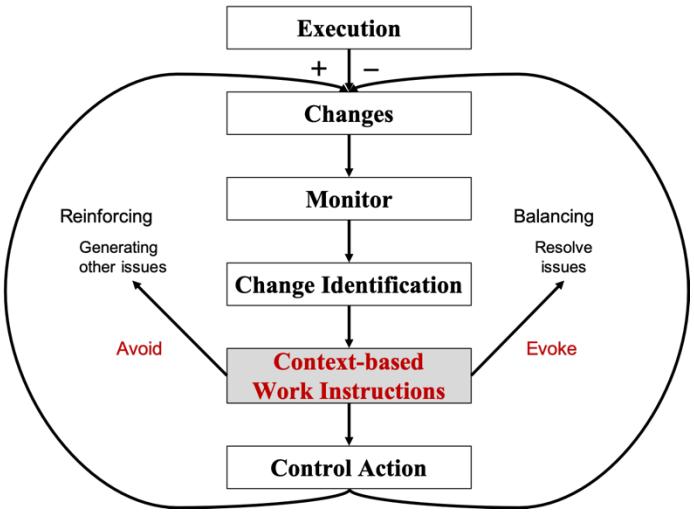

Figure 15. Revised feedback processes caused by changes (Motawa et al., 2007)

4.3 Just-in-time principles and Last Planner System

Just-in-time (JIT) is a core feature of the lean philosophy that originated from the Toyota Production system which was originally designed for automotive manufacturing (Golhar and Stamm, 1991). JIT uses a pull mechanism to convey the needs of replenishing parts from a downstream process to a preceding process. The ultimate goal of JIT is to ensure that each process gets the right parts at the right time and quantity. The concept of JIT was later incorporated into the construction industry to achieve lean construction (Ballard and

Howell 1995; Tommelein 1997; Tommelein 1998). Physical objects are always attached with information, and, as such, the flow of physical objects usually aligns with the information flow (Fujimoto, 1999). JIT is related to this research in that the information flow should be bi-directional, which means that crew members should not only be passively given information but also actively provide information in line with the pull mechanism. As mentioned at the beginning of this paper, work instructions are the media to transmit information on construction jobsites; however, paper-based work instructions are not able to communicate information in a just-in-time way. As a result, field crews could be confronted with an information deficit, which may lead to on-site decisions being deferred. Since this research is intended to implement the JIT concept in field communication, mobile-based context-aware applications are a promising technology. The benefits of doing so is that changes of the field context will be considered and be used to update the context-based work instructions in a just-in-time manner.

In line with the JIT concept, Ballard (2000) designed the Last Planner System (LPS) to produce predictable workflow on construction jobsites. The implementation of the LPS involves "Should-Can-Will-Did" planning and feedback processes that correspond to five parts (i.e., master planning, phase planning, make-ready planning, weekly work planning, and learning) as shown in Figure 16. The "Should" process identifies what should be done to successfully complete a project, and the planning takes place at least months ahead of time. At this stage, the context information is mainly about the design, schedule, specifications, etc. Meanwhile, the context information at this stage may be too broad for field crews. Moreover, the information usually is based on assumptions that need further verification before use. Moving to the "Can-Will-Do" processes, the planning is brought to the weekly level. Now, the last planners (superintendents and foremen) are more actively involved in the planning processes. The "push" mechanism passes down context information from different context sources (e.g., design information is sourced from designers); at the same time, though, the "pull" mechanism starts to impact the information flow. In this research, we propose that the communication of context-based work instructions could align with the "Can-Will-Do" processes on a weekly basis. After the weekly planning is made, context information is collected to create work instructions for

field crews. Since the information is still assumed, the last planners should be cautious and further verification of the context is needed. Especially, when there are changes of field context passed down from context sources or reported by field crews, context-based work instructions should be updated. The verification and update of the field context should be as-needed in that it could be daily or even real-time (e.g., weather). Figure 16 depicts how the context-based work instructions would be prepared and updated along with the LPS processes.

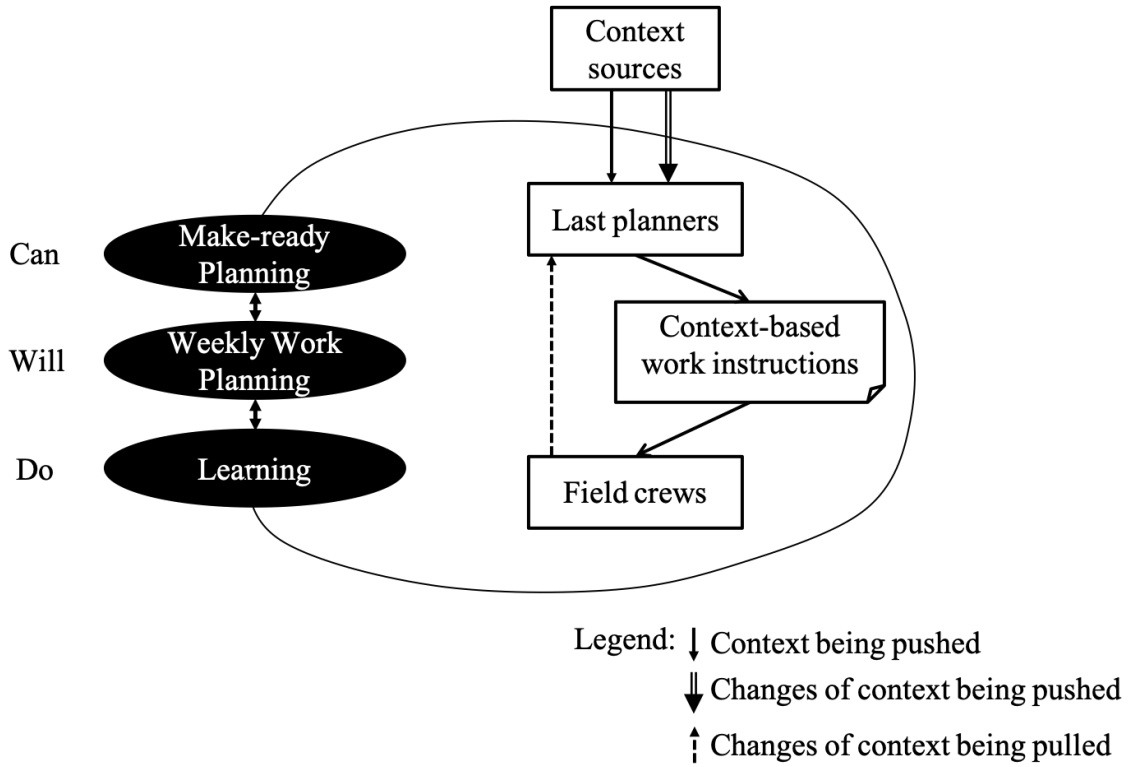

Figure 16. Preparation and update of context-based work instruction in line with the Last Planner System processes

4.4 Context-aware applications on construction jobsites

In the construction domain, many researchers have attempted to implement enabling technologies to provide context-aware services for field crews, which indicates that the importance of context has been realized. Based on the synthesis of previous definitions of context and characteristics of construction field tasks, we put forward a definition of the

field context as "up-to-date information about the situation of construction field crews that could influence the process or outcome of field tasks within a certain time frame". In order to provide context information for field crews, it is necessary to have a relatively complete list of context types. In a prior study, we have identified and validated 17 basic field context types: "design", "material", "component", "crew composition", "crew member", "task", "schedule", "equipment and tools", "weather", "work zone", "site condition", "nearby objects", "technical and safety specifications", "cultural and legal constraints", "industry policy", "public relation", and "computing entity". With an intention to leverage mobile computing technologies to communicate context-based work instructions, we refer to previous studies to figure out how a context-aware application could be developed.

Singhvi and Terk (2003) developed a context information system framework called Prophet to proactively identify and retrieve information for workers on construction jobsites. They defined context as "any information that helps in the dynamic characterization of the construction environment. The basic elements of context are identity of the user, temporal information, location and community". However, the definition of context proposed in the research is not specific enough to guide field crew members to perform tasks. Lee and Mohsen (2008) developed a construction context classification and a Context Aware Computing (CAC) system to deliver appropriate information that can help workers deal with the dynamics of construction. In their research, the construction context classification includes work item context, user context, and environment context. However, the classification does not consider how the work is performed. Wang et al. (2011) define context as "situations in which a construction entity acts or exists; whereby an entity can be a person involved in the construction, an action taken to complete a job, a component built on a project, or a resource utilized to perform the action or build the component". Further, they presented a framework based on the notion of context descriptors. The framework is made up of two components: a representation of context descriptors using OWL-based ontologies and a reasoning mechanism to determine new descriptors' applicability when the number of context descriptors increases. Lee et al. (2013) developed a framework for Information Lifecycle Management (ILM) using RFID to help material control. In this research, they considered context as "on-going variables

resulting from a given environment as well as a variety of participants, equipment, and scheduled works". Context parameters are identified as RFID checkpoint, user, time, location, and lifecycle stage. In these studies, the researchers first attempted to define context and identify specific types of context. However, the listed context types may fall short of context information for construction field crews.

Furthermore, there were studies focused on enabling technologies. Aziz et al. (2005) reviewed enabling technologies for intelligent wireless web services to help field crews access context-relevant and personalized information. Anumba and Aziz (2006) developed a context-aware service delivery architecture which has three major functions: context capture, context inference and context integration. With the architecture, context information is captured from sources including user's current location, user device type, user identity, user's current activity, visual context via a CCTV-over-IP camera and time via a computer lock. Thereafter, the context inference tier reasons about the captured context, retrieves relevant information and, lastly, the context integration tier integrates the captured and inferred context to trigger pre-programmed events. Rebolj et al. (2008) argue that reliable context information is critical for successful completion of tasks, and they propose a method consisting of three components: an automated activity tracking subsystem based on image recognition, an automated material tracking subsystem, and a mobile computing supported communication environment. Instead of exploring the use of certain mobile computing technologies, Chen and Kamara (2011) introduce a framework consisting of an application model and a technical model: the application model identifies the features of mobile computing, construction personnel, construction information, and construction sites, and they explore the interactions affecting the implementation of mobile computing; the technical model presents a clear structure for designing mobile computing systems from a technical perspective.

These studies were devoted to exploring technologies that could be leveraged to provide context information for field crews. They led us to the conclusion that a context-aware system should at least consist of two components: collection of context information and communication. We decided to attempt to develop a minimum viable prototype (MVP) to

help collect context information, prepare context-based work instructions, and communicate instructions to field crews.

4.5 Summary

Given the above discussions, state-of-the-art field work instructions cannot respond to changes of context information or get updated with the changed context in a timely manner. Moreover, researchers have attempted to develop context-aware applications for construction field personnel, but none has been devoted to providing field crews with context information and its changes. Hence, this research aims to create context-based work instructions that are informed of changes and can be communicated in a just-in-time way to enhance context-awareness for construction field crews.

5. Research method

The research method was designed to include three major steps: (1) design a method to collect context information for field crews and (2) develop context-based work instructions on a mobile-based prototype. The mobile-based prototype serves as an MVP in this study. It helps to facilitate communication of context-based work instructions; and (3) conduct a field study to evaluate whether context-based work instructions impact the cycle time variability of field tasks.

5.1 Collection of the field context

The proposed context-based work instructions were developed based on the 17 basic types of context information. The first step was to determine the sources of the information. To simplify, we divided the information into two broad categories: project-specific information and information generally applicable to construction. As for a construction company, project-specific information is only applicable to a certain project, while information generally applicable to construction can be applied to every project. Figure 17 shows how each context type is sourced from different participating actors.

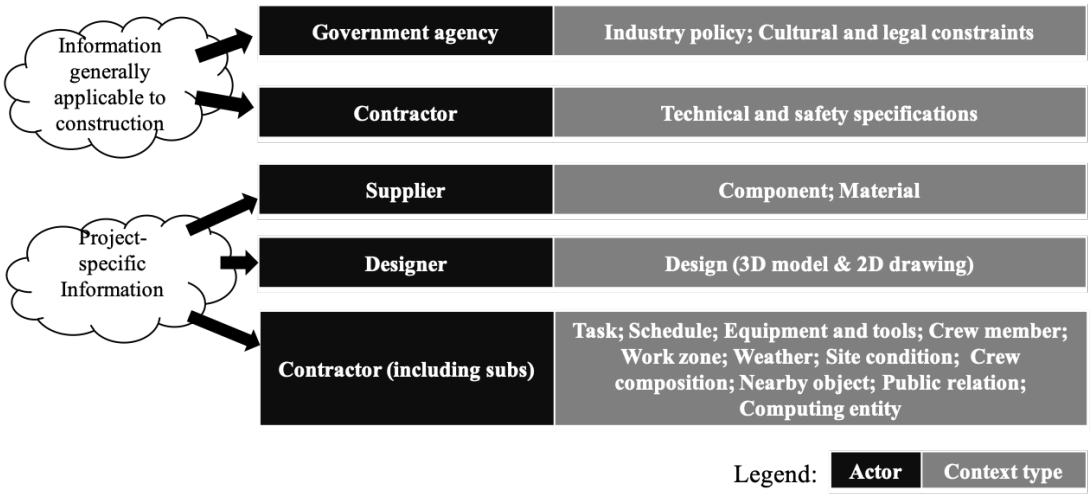

Figure 17. Sources for the 17 context types

For example, a subcontractor can provide information about equipment and tools specific to a certain task, while the industry policy should come from the relevant government agency. Once each type of context information is obtained for a task, a work instruction can be created on the mobile-based prototype. In this research, the context information was acquired manually by the first author who was on site and worked closely with the field crews.

5.2 Context-based work instruction prototype (MVP)

To provide context-based work instructions for field crews, a mobile-based prototype was created to leverage mobile computing technologies. The mobile-based prototype was developed to meet the criteria that context-based work instructions can be informed of changes and be communicated in a just-in-time way.

The implementation of context-based work instructions is designed to include three stages: preparation, distribution, and update. In the preparation stage, the context information is collected to prepare context-based work instructions. We divided the 17 context types into two batches because the time required to collect the information and respond to changes is different. The first batch includes “task”, “schedule”, “design”, “crew composition”, “crew

member", "material", "component", "equipment and tool", "tech and safety specification", "cultural and legal constraints", "public relation", "industry policy" and "computing entity"; the second batch includes "weather", "work zone", "site condition", and "nearby object". Context information in the first batch should be collected a week before the task starts, and the second batch should be checked a day ahead. In the distribution stage, work instructions will be sent to field crews for review (at least thirty minutes) before their task begins. Crew members are then requested to review the instructions. In the update stage, work instructions will be updated by our onsite research staff when there are changes of context information.

Throughout the stages, field crews should check whether the predefined context remains valid; for instance, the delivery of a precast wall may be delayed, in which case, the superintendent and foreman need to make an adjustment to their original plan. Any change to the predefined context will be considered by our onsite research staff to update the work instruction and field crews will get a notification of the change on the mobile-based prototype.

5.3 Conduct a field study

We conducted a field study to evaluate the impact of context-based work instructions on the variability of task cycle time in two dimensions: (1) comparison between the actual and the planned cycle time; and (2) the variability of the actual cycle time for the same type of field tasks. The field study was conducted on an office renovation project located in China.

The field study lasted for two months, during which context-based work instructions were developed and provided for field crew members using the mobile-based prototype. A total of 246 instances falling in 14 groups (i.e., the same type) of field tasks were observed in the field study. For the same group of field tasks, a control sub-group and a treatment sub-group were set up. Context-based work instructions were developed, distributed, and updated using the mobile-based prototype for the treatment sub-group, while state-of-the-art work instructions were provided for the control group (Mourgues and Fischer, 2008). We recorded time-lapse videos of these 246 instances of field tasks and then extracted three

types of information: start time, end time, and context-related problems associated with each instance of task.

In line with (1), for each task in the treatment and control sub-group, we compared the actual task cycle time with the planned, and the delta was calculated using the following formula.

$$\text{Cycle time} = \text{Finish time} - \text{Start time} \tag{1}$$

$$\text{Delta} = \text{Actual cycle time} - \text{Planned cycle time} \tag{2}$$

Thereafter, a percent deviation (PD) was calculated using the following formula to determine by how much the actual cycle time differed from the planned cycle time.

$$\text{Percent deviation} = \text{Delta} / \text{Planned cycle time} \tag{3}$$

Knowing the percent deviation of the treatment and control sub-group, we then calculated the difference between the two percent deviations. The percent deviation (PD) difference indicates whether the availability of updated context information impacts the ability of the crew to meet the planned cycle time and whether the variability of cycle times for the same type of task is higher or lower.

$$\text{PD Difference} = \text{Control group PD} - \text{Treatment group PD} \tag{4}$$

In line with (2), we evaluated the variability of the actual cycle times for both control and treatment sub-groups in 6 out of these 14 groups of field tasks. Tasks for the selected six groups were installations of the steel frame wall and wood plates. We selected these 6 groups because we were able to calculate the cycle time per quantity (e.g., meter, square meter) so that we can maintain the consistency for our evaluation. The metric we calculated for the evaluation is the variance of the actual cycle time per quantity in these groups.

6. Field study results

Figure 18 shows the averaged percent deviation within each control and treatment sub-group in the 14 groups of field tasks. Detailed calculations of the deltas and percent deviation differences for each sub-group are shown in Table 8 in the Appendix section. The delta of the control group itself indicates that the cycle time of these field tasks went awry from the original plan. The field crews in the control sub-group were provided with state-of-the-art work instructions which did not allow updates of work instructions due to changes in context and did not communicate the information in a timely manner. As a result, the execution of field tasks could be affected by occurrences of unexpected changes of context, of which field crews were not aware in the control sub-groups.

As to the first dimension of evaluation, we found that, for each group, the averaged delta in the treatment sub-group was generally less than that in the control group. For instance, the installation of the ceiling board in the control sub-group has an averaged delta of 71.40%, i.e., the actual cycle time for the installation was much more than the planned cycle time. The reasons accounting for the large delta were that field crews were confronted with situations including unexpected occupations of the work zone, shortage of boards, unavailability of tools, and unclear height. For the corresponding treatment sub-group, with the implementation of context-based work instructions, field crews received updates of the context ahead of time and hence could adjust their tasks accordingly. For instance, for the installation of steel frame walls, field crews were receiving timely updates mostly on the design (e.g., the length of the frame wall is incorrect), material (e.g., steel does not meet specification), work zone (e.g., another crew occupies the zone) and schedule (conflicts of schedule). Overall, the delta of the treatment group is within 10%. On average, the percent deviations of the 14 treatment sub-groups is 35.7% less than that of the 14 control sub-groups. The percent deviation difference, represented using the grey bar in the figure, indicates that the cycle time of the treatment sub-groups was subject to much less variability than the control sub-groups.

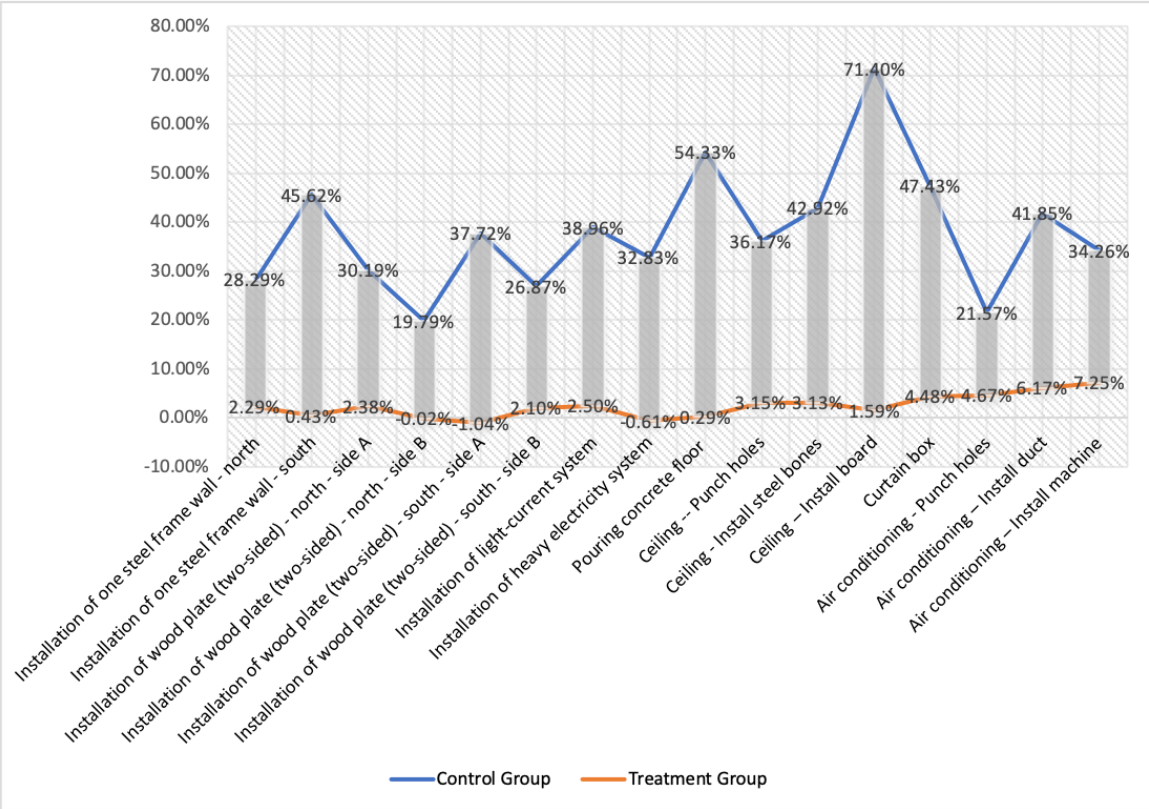

Figure 18. Percent deviation of the control and treatment sub-groups

As to the second dimension of evaluation, we found that the variance of the actual cycle time per quantity in the treatment sub-groups was generally smaller than in the control group. This implies that, with context-based work instructions, the actual cycle time varied less in the treatment sub-groups. Our calculations results are shown in Table 7.

Table 7. Variance of the actual cycle time per quantity in the control and treatment sub-groups

Task	Variance of the actual cycle time per quantity in the control sub-group	Variance of the actual cycle time per quantity in the treatment sub-group
Installation of one steel frame wall – north (quantity: meter)	0.0262	0.0010
Installation of one steel frame wall – south (quantity: meter)	0.1111	0.0035
Installation of wood plate (two-sided) - north - side A (quantity: square meter)	0.1838	0.0028
Installation of wood plate (two-sided) - north - side B (quantity: square meter)	0.0531	0.0071
Installation of wood plate (two-sided) - south - side A (quantity: square meter)	0.1260	0.0093
Installation of wood plate (two-sided) - south - side B (quantity: square meter)	0.0583	0.0217

7. Discussion

With the intention to inform construction field crews of context information and its changes in a timely manner, this study provides context-based work instructions for construction field crews. To evaluate the impact of context-based work instructions on the cycle time of field tasks, we designed a controlled experiment as shown in section 5.3. In the discussion section, we reflect on the medium for communicating context-based work instructions and the impact of context-based work instructions on production control.

1) Medium for communicating context-based work instructions

Mourgues and Fischer (2008) developed a field instruction template and they provided paper-based instructions. Following their way, we started to provide a concrete crew with paper-based instructions that incorporate context information on a tested site. However, it turned out not working because it was difficult to inform field crews of changes of context. In fact, paper-based instructions make it difficult to exchange information on jobsites (Bowden et al., 2004). It should also be noted that Mourgues and Fischer (2008) did not consider changes in their study.

Meanwhile, we noticed that the use of mobile devices is on the upswing on construction sites. More affordable mobile devices along with increased wireless network speeds provide opportunities for enhancing on-site communication (Chen and Kamara, 2008; Sattineni and Schmidt, 2015). For the ease of context information exchange, therefore, we deployed a mobile-based prototype to communicate context-based work instructions in the case study. Specifically, we chose tablets to be the device for communication. However, this study does not compare the advantages and/or disadvantages of every type of available mobile devices (e.g., smart phones, augmented reality glasses) (Nourbakhsh et al., 2012).

2) Impact of context-based work instructions on production control

Construction production control has been challenging. Traditionally, ad-hoc methods for production control are often informal and ineffective in mitigating variability. To tackle this problem, lean construction methods (e.g., the Last Planner System) suggest involving field crews in the decision-making process and getting reports of site information from them (Ballard 1997; Ballard 1999). However, the inherently dynamic production may require shorter planning cycles (e.g., daily) than normal lookahead planning cycles (e.g., weekly). Dave et al. (2016) discussed the importance of lean and close to real-time reporting of production control information, and they mentioned the challenge to inform field crews of such information in a timely manner. This study provides a new form of work instructions to inform field crews of context information and its changes in a timely manner. With context-based work instructions on a mobile-based prototype, information of certain context types is checked on a daily basis. Instructions get updated when there is any change of context. As indicated in section 6, this paper shows that context-based work instructions are promising in helping mitigate the variability of task cycle time on construction sites. Therefore, context-based work instructions could become a measure for construction production control. Meanwhile, we realize that there could be challenges for implementing the proposed context-based work instructions, including: information overload and lack of access to mobile devices (Dave et al., 2016).

8. Conclusions

Providing construction field crews with high-quality work instructions that could inform field crews of context and its changes in a timely manner has been an elusive goal. This paper presents an innovative effort towards developing context-based work instructions for construction field crews. We extended criteria on work instructions by focusing on two features: informed of changes and just-in-time. The first feature is important because field tasks are often subject to changes in which case field crews would not successfully complete their tasks if the instructions given to them do not consider the changed context. The context-based work instructions will help field crews manage changes on construction jobsites in that balancing feedback processes will be evoked to resolve issues. As to the second feature, the preparation, distribution, and update of work instructions were designed to align with short-cycle planning processes on construction jobsites. Moreover, a mobile-based prototype was developed as an MVP to facilitate communication of context-based work instructions. As a result, the proposed work instructions could inform field crews of the context information and its changes in a just-in-time way.

Through performing a case study on an office renovation project, we evaluated how context-based work instructions impact the cycle time of construction field tasks. For the same type of field task, the percent deviation of the treatment sub-group is much smaller than that of the control group. A smaller percent deviation in the treatment sub-group means that the actual cycle time is closer to the planned cycle time. Overall, the difference between the percent deviation for the 14 control sub-groups and the 14 treatment sub-groups was averaged to 35.7%. The result indicates that, with the implementation of context-based work instructions, the actual cycle time was more aligned with the planned. Moreover, it was found that the cycle times of the same type of field tasks vary less. Therefore, it can be concluded that the variability of the cycle time of field tasks was reduced in the treatment group. Furthermore, the percent deviation difference also implies that the actual cycle time was 35.7% less than the planned cycle time on average attributed to the implementation of context-based work instructions. This time saving from the

application of context-based work instructions could be more significant as it would accumulate if context-based work instructions were applied to more tasks for a project.

Overall, this research provides a new form of work instructions based on the recognition of field context. The proposed work instructions are communicated in line with short-cycle planning processes in that these three stages (i.e., preparation, distribution, and update) of work instructions are in line with weekly planning. The majority of the 17 field context types for a task should be collected and made ready for use a week before the task starts. The update of work instructions is designed to be as needed since changes could happen unexpectedly; however, the instructions could help to mitigate the negative impacts of changes. Moreover, the "push-pull" mechanism is applied to the three stages of work instructions so that field crews are empowered to provide feedback to predefined context.

This research has the following limitations. First, due to the lack of accessibility to construction jobsites during the Covid-19 pandemic, we only conducted the case study on one project and 246 instances of field tasks. The generalizability of the case study result may be criticized. Therefore, we will extend the evaluation of the proposed context-based work instructions to more projects and types of field tasks. It would be interesting to look into whether the context-based work instructions could help reduce the variability of the cycle time for other tasks on other projects. Moreover, the impact of context-based work instructions may not be just on the cycle time. We are curious about what other benefits the instructions could bring to construction field crews. For example, we would explore whether context-based work instructions make construction jobsites a safer work place for field crews. Besides, to prepare the instructions, the last planners should collect the field context information. The weather context could be sensed automatically, whereas most of the other context types were manually collected and then input into the mobile-based prototype. Future work will explore whether each context types could be automatically captured and how, which would make the preparation of instructions much easier. Lastly, future work may explore what mobile devices could be leveraged to facilitate communication of context-based work instructions. For instance, augmented-reality glasses could be promising.

Acknowledgement

This study was funded by the Seed Funding provided by the Center for Integrated Facility Engineering (CIFE) at Stanford University.

Appendix

Table 8. Calculations of percent deviation and its difference for the control and treatment sub-groups

Task	Percent deviation (PD) between the planned and the actual cycle time		PD Difference
	Control sub-group	Treatment sub-group	
Installation of one steel frame wall - north	28.29%	2.29%	25.99%
Installation of one steel frame wall - south	45.62%	0.43%	45.19%
Installation of wood plate - north - side A	30.19%	2.38%	27.81%
Installation of wood plate – north - side B	19.79%	-0.02%	19.80%
Installation of wood plate - south - side A	37.72%	-1.04%	38.75%
Installation of wood plate - south - side B	26.87%	2.10%	24.77%
Installation of light-current system	38.96%	2.50%	36.46%
Installation of heavy electricity system	32.83%	-0.61%	33.43%
Pouring concrete floor	54.33%	0.29%	54.03%
Ceiling - Punch holes	36.17%	3.15%	33.02%
Ceiling - Install steel bones	42.92%	3.13%	39.79%
Ceiling - Install board	71.40%	1.59%	69.81%
Installation of curtain box	47.43%	4.48%	42.95%
Air conditioning - Punch holes	21.57%	4.67%	16.90%
Air conditioning - Install duct	41.85%	6.17%	35.69%
Air conditioning - Install machine	34.26%	7.25%	27.01%

References

Antifakos, S., Michahelles, F. and Schiele, B., 2002, September. Proactive instructions for furniture assembly. In *International Conference on Ubiquitous Computing* (pp. 351-360). Springer, Berlin, Heidelberg.

Anumba, C. and Aziz, Z., 2006, June. Case studies of intelligent context-aware services delivery in AEC/FM. In *Workshop of the European Group for Intelligent Computing in Engineering* (pp. 23-31). Springer, Berlin, Heidelberg.

Aziz, Z., Anumba, C.J., Ruikar, D., Carrillo, P.M. and Bouchlaghem, D.N., 2005, July. Context aware information delivery for on-site construction operations. In *Proceedings of the 22nd CIB-W78 Conference on Information Technology in Construction, Institute for Construction Informatics, Technische Universitat Dresden, Germany, CBI Publication* (Vol. 304, pp. 321-332).

Austin, P.E., Matlack, R., Dunn, K.A., Kesler, C. and Brown, C.K., 1995. Discharge instructions: do illustrations help our patients understand them?. *Annals of emergency medicine*, *25*(3), pp.317-320.

Ballard, G. and Howell, G., 1995. Toward construction JIT. *Lean construction*, *291*, p.300.

Ballard, G., 1997, July. Lookahead planning: the missing link in production control. In *Proc. 5th Annl. Conf. Intl. Group for Lean Constr*.

Ballard, G., 1999, July. Improving work flow reliability. In *Proceedings of the 7th Annual Conference of the International Group for Lean Construction* (pp. 275-286).

Ballard, H.G., 2000. *The last planner system of production control* (Doctoral dissertation, University of Birmingham).

Bowden, S., A. Dorr, A. Thorpe, and C. J. Anumba. "Mapping site processes for the introduction of mobile IT." In *ECPPM*, vol. 4, pp. 491-8. 2004.

Chen, Y. and Kamara, J., 2008, June. The mechanisms of information communication on construction sites. In Forum Ejournal (Vol. 8, No. 1, pp. 1-32).

Chen, Y. and Kamara, J.M., 2011. A framework for using mobile computing for information management on construction sites. *Automation in construction*, *20*(7), pp.776-788.

Dave, Bhargav, Sylvain Kubler, Kary Främling, and Lauri Koskela. "Opportunities for enhanced lean construction management using Internet of Things standards." *Automation in construction 61* (2016): 86-97.

Di Eugenio, B., 1992, June. Understanding natural language instructions: the case of purpose clauses. In *30th Annual Meeting of the Association for Computational Linguistics* (pp. 120-127).

Fujimoto, T., 1999. *The evolution of a manufacturing system at Toyota*. Oxford university press.

Gao, J. and Fischer, M., 2006. Case studies on the implementation and impacts of virtual design and construction (VDC) in Finland. *Center for Integrated Facility Engineering, Stanford University*.

Golhar, D.Y. and Stamm, C.L., 1991. The just-in-time philosophy: a literature review. *The International Journal of Production Research*, *29*(4), pp.657-676.

Gorse, C.A. and Emmitt, S., 2003. Investigating interpersonal communication during construction progress meetings: challenges and opportunities. *Engineering, Construction and Architectural Management*.

Haymaker, J., Ackermann, E. and Fischer, M., 2000. Meaning mediating mechanism. In *Artificial Intelligence in Design'00* (pp. 691-715). Springer, Dordrecht.

Heiser, J., Phan, D., Agrawala, M., Tversky, B. and Hanrahan, P., 2004, May. Identification and validation of cognitive design principles for automated

generation of assembly instructions. In *Proceedings of the working conference on Advanced Visual Interfaces* (pp. 311-319).

Koo, B. and Fischer, M., 2000. Feasibility study of 4D CAD in commercial construction. *Journal of construction engineering and management*, *126*(4), pp.251-260.

Lee, J.H., Song, J.H., Oh, K.S. and Gu, N., 2013. Information lifecycle management with RFID for material control on construction sites. *Advanced engineering informatics*, *27*(1), pp.108-119.

Lee, S. and Mohsen, O., 2008, June. Application of Context-Aware Computing to Construction Field Work. In *Proceedings of the CSCE 2008 Annual Conference* (Vol. 1, No. 2, pp. 483-490).

LeFevre, J.A. and Dixon, P., 1986. Do written instructions need examples? *Cognition and Instruction*, *3*(1), pp.1-30.

Li, W., Ritter, L., Agrawala, M., Curless, B. and Salesin, D., 2007. Interactive cutaway illustrations of complex 3d models. *ACM Transactions on Graphics (TOG), 26*(3), pp.31-es.

Li, X., Liu, Z. and Jifeng, H., 2004, April. A formal semantics of UML sequence diagram. In *2004 Australian Software Engineering Conference. Proceedings*. (pp. 168-177). IEEE.

Motawa, I.A., Anumba, C.J., Lee, S. and Peña-Mora, F., 2007. An integrated system for change management in construction. *Automation in construction*, *16*(3), pp.368-377.

Mourgues, C. and Fischer, M., 2008. A work instruction template for cast-in-place concrete construction laborers. *Center for Integrated Facility Engineering. Stanford University. Working Paper*, *109*.

Nasir, D., McCabe, B. and Hartono, L., 2003. Evaluating risk in construction–schedule model (ERIC–S): construction schedule risk model. *Journal of construction engineering and management*, *129*(5), pp.518-527.

Nourbakhsh, Mehdi, Rosli Mohamad Zin, Javier Irizarry, Samaneh Zolfagharian, and Masoud Gheisari. "Mobile application prototype for on-site information management in construction industry." Engineering, construction and architectural management (2012).

Othman, A.A., Hassan, T.M. and Pasquire, C.L., 2004. Drivers for dynamic brief development in construction. *Engineering, Construction and Architectural Management*.

Rebolj, D., Babič, N.Č., Magdič, A., Podbreznik, P. and Pšunder, M., 2008. Automated construction activity monitoring system. *Advanced engineering informatics*, *22*(4), pp.493-503.

Sattineni, Anoop, and Taylor Schmidt. "Implementation of mobile devices on jobsites in the construction industry." *Procedia Engineering 123* (2015): 488-495.

Singhvi, V., Fish, W. and Terk, M., 2003. Context-aware information system for construction site applications. In *Construction Research Congress: Wind of Change: Integration and Innovation* (pp. 1-8).

Sterman, J.D., 1992. System dynamics modeling for project management. *Unpublished manuscript, Cambridge, MA*, *246*.

Tommelein, I.D., 1997, July. Discrete-event simulation of lean construction processes. In *Proceedings of the Fifth Conference of the International Group for Lean Construction*.

Tommelein, I.D., 1998. Pull-driven scheduling for pipe-spool installation: Simulation of lean construction technique. *Journal of construction engineering and management*, *124*(4), pp.279-288.

Wang, H., Boukamp, F., and Elghamrawy, T. (2011). “Ontology-based approach to context representation and reasoning for managing context-sensitive construction information.” *Journal of Computing in Civil Engineering*, *25*(5), 331–346.

Chapter 5: Conclusion

Effective field communication is critical to ensure that field crews obtain the necessary information to perform their tasks on construction jobsites. However, lacking understanding of context and its changes has eluded construction field crews and has led to unneglectable variations in task cycle time. Analysis of state-of-the-art work instructions for field crews showed that current instructions do not inform crews sufficiently of context and its changes in a timely manner.

To address the above problem, this research explored how context could be leveraged to improve work instructions for field crews. The first step was to have a clear understanding of the context for construction field crews and develop a context taxonomy to provide a common vocabulary. Moreover, I developed work instructions by incorporating context information and assess the impact of context-based work instructions on the cycle time of field tasks. Details of each research task can be found in Chapter 2 through 4. In Chapter 1, this thesis presents an overview of the theoretical contributions to knowledge. In this chapter, I discuss the theoretical contributions at a more detailed level and their limitations, the validation, and practical implications. I conclude this thesis with suggestions for future work.

1. Theoretical contributions

This thesis has the following theoretical contributions:

- An adjusted definition of context and a context taxonomy consisting of 17 context types for construction field crews, and
- Assessment of the impact of context-based work instructions on the cycle time of construction field tasks.

I explain these contributes in detail and discuss their limitations below.

Chapters 2 and 3 correspond to the first contribution. Although researchers have attempted to improve context awareness on construction jobsites, we still lack a clear understanding of context and general context types for field crews. This research fills this gap by adjusting previous definitions of context and put forward one that is specific to construction field crews. The adjusted definition of field context information is "up-to-date information about the situation of construction field crews that could influence the process or outcome of field tasks within a certain time frame". This definition emphasizes two features of context. The first feature is "timeliness", which corresponds to "within a certain time frame", and the second feature is "relevancy", which corresponds to "influence the process or outcome of field tasks". Based on the adjusted definition, preliminary field studies and existing construction information schemas, including the POP, AROW and AFM models, I identified an initial list of 17 context types. This initial list of context types was validated through semi-structured interviews with 15 experienced industry professionals from 5 sophisticated construction companies. These interviewees had abundant experience working with construction field crews for about 12 years on average. Their opinions yielded convincing feedback on the significance of each context type for construction field crews. Moreover, I also asked each interviewee whether they could think of another context type which was not included in the initial list and no additional context type was reported. Hence, this thesis shows that the initial list forms a complete list of context types for construction field crews.

Based on the context taxonomy, I further built a Construction Context Information (CCI) model in the form of an OWL-encoded ontology. The context model defines relationships between these 17 context types and their attributes. The CCI model enhances the ability to interpret, communicate and reason about field context information. Specifically, reasoning rules were defined to show that inferences of high-level context (e.g., a task's status) could be performed based on low-level context (i.e., context that could be directly defined or sensed). The CCI model could be the basis for developing context-aware systems for construction field crews.

Chapter 4 corresponds to my second contribution. I assessed the impact of context-based work instructions on the cycle time of field tasks. The assessment was performed through a case study on an office renovation project. More specifically, I designed a controlled experiment on 14 pairs of control and treatment groups. Each pair was made up of tasks of the same type. For quantitative evaluation, I used the cycle time of tasks as the basic metric. Thereafter, the delta between the actual cycle time and the planned cycle time was calculated to show the reliability of planning. To make the evaluation consistent across these 14 pairs, I further calculated the percent deviation as the delta over the planned cycle time. Details of the quantitative assessment can be found in Chapter 4. The controlled experiment showed that the actual cycle time was more consistent with the planned cycle time of field tasks in the treatment group than in the control group. In general, the delta of field tasks in the treatment group was shown to be less than that in the control group. On average, the percent deviation of the treatment group was 35.7% less than that of the control group. It indicates an improved alignment between the actual and planned cycle time. Moreover, the actual cycle times of the same type of field tasks varied less in the treatment group than in the control group. Overall the quantitative analysis implied that the context-based work instructions could reduce the variability of the cycle time of field tasks.

2. Practical contributions

This thesis contributes context-based work instructions that could be implemented in line with short-cycle planning for construction field crews. Detailed description of how context-based work instructions could be used are present in Chapter 4. The implementation requires involvement of field crews, including superintendents, foremen, field engineers and crew members. A three-stage implementation process was proposed, including preparation, distribution, and update of work instructions. The case study shows that it is promising that context-based work instructions could help to communicate the field context information and inform crews of changes in a timely manner.

Moreover, this thesis provides a mobile-based solution that facilitates communication of the field context information. In practice, oral or paper instructions are still the most popular. Mobile-based solution could be a substitute. The mobile-based prototype provides

a way to quickly provide and update the field context information for construction field crews.

Furthermore, context-based work instructions reduce variability in the cycle time of field tasks. When informed of context information and its changes in a timely manner, field crews were able to avoid using outdated information while performing tasks. As shown in the case study, a higher consistency between the actual cycle time and the planned cycle time of field tasks was observed in the treatment group. Moving up from the task level to the project level, context-based work instructions could be leveraged for better production control and could help prevent schedule overruns.

Especially for change management, context-based work instructions provide a more proactive way to deal with changes. In case of changes, instructions get updated and field crews will be informed. Superintendents and foremen could therefore make more informed decisions to avoid or mitigate negative impacts of changes.

More broadly, context-based work instructions could be used as records of field task information. The instructions keep a track of what is planned, changed, and completed for field tasks. When different field crews run into disputes, the instructions could provide a proof of how the tasks are performed and, hence, may help resolve disputes.

3. Limitations

This thesis has the following limitations:

First, the validation of these 17 context types was performed through semi-structured interviews with 15 experienced industry professionals. Although these interviewees have worked in the construction industry for about 12 years on average and they worked closely with field crews, the number of interviewees was limited. If more industry professionals were interviewed, the results could be more convincing.

Moreover, the interview answers indicated that the importance of these context types may vary for different types of field tasks. However, the interviews did not provide sufficient data allowing for an analysis of how important each context type could be, or how the importance could vary for different types of field tasks.

Built upon the 17 context types, an ontology-based context model is presented in this thesis. The model was built for a stronger expressiveness by relating the context types and defining attributes of each context type. However, this thesis does not validate these relationships and attributes. As a continuing effort, the expert interview method could be applied to validate these relationships and attributes defined in the context information model.

Regarding the implementation of context-based work instructions, only one case study was conducted due to the limited access to construction jobsites. The Covid-19 pandemic made it more difficult to get onto a construction jobsite. Moreover, the analysis was performed based on data collected from 14 types of tasks on an office renovation project. The generality of the analysis might be criticized for the limited number of case study and types of field tasks. In addition, the impact was evaluated on the cycle time of field tasks. This thesis did not consider other quantitative metrics (e.g., the number of workface questions) because of the difficulty in collecting such metrics.

4. Future work

According to the conceptual research roadmap I showed in Chapter 1, this thesis presents the identification of a context taxonomy consisting of 17 specific context types and context-based work instructions for construction field crews. For researchers who are interested in enhancing context-awareness on construction jobsites, this thesis lays the theoretical foundation. This thesis could lead to further explorations in the following areas.

Field communication is characterized with a high volume of information. The collection of context information in the preparation stage of work instructions might require a lot of effort from the field team without intelligent data collections methods. Hence, one

promising area of study could be on how the field context information could be collected automatically. Researchers should explore feasible solutions (e.g., deployment of sensors) to automatic collection of context information on construction jobsites. Solutions may be assessed in terms of technological feasibility, money investment, and return on investment.

In terms of how context-based work instructions could be communicated to field crews, this thesis presents a solution in the form of a web-based application. Field crews were able to view the instructions using a smartphone or a tablet. There are other mobile devices that have the potential to be implemented. For instance, an augmented-reality glass could be a possible solution. Future research should compare the pros and cons of these mobile devices.

Moreover, the impact of context-based work instructions was mainly evaluated on the cycle time of field tasks. The impact could also be assessed on other areas. For instance, safety is a big concern for field crews. Context-based work instructions could inform field crews of possible safety hazards and, therefore, help avoid safety incidents. Hence, future work should evaluate how context-based work instructions impact the safety and other types of performance on construction jobsites.

ProQuest Number: 29003853

INFORMATION TO ALL USERS
The quality and completeness of this reproduction is dependent on the quality
and completeness of the copy made available to ProQuest.

Distributed by ProQuest LLC (2022).
Copyright of the Dissertation is held by the Author unless otherwise noted.

This work may be used in accordance with the terms of the Creative Commons license
or other rights statement, as indicated in the copyright statement or in the metadata
associated with this work. Unless otherwise specified in the copyright statement
or the metadata, all rights are reserved by the copyright holder.

This work is protected against unauthorized copying under Title 17,
United States Code and other applicable copyright laws.

Microform Edition where available © ProQuest LLC. No reproduction or digitization
of the Microform Edition is authorized without permission of ProQuest LLC.

ProQuest LLC
789 East Eisenhower Parkway
P.O. Box 1346
Ann Arbor, MI 48106 - 1346 USA